

(a) (b) (c)

Figure 22.6 Abnormally shaped red blood cells help make the diagnosis in haemolytic anaemias. **(a)** Spherocytes (arrows) in hereditary spherocytosis. **(b)** Sickle cells (arrows) in sickle cell disease. **(c)** Hypochromic cells (arrows) in thalassaemia.

gene mutations in ribosomal protein (RPS) genes are implicated in some cases. Most cases present at 2–3 months of age, but 25% present at birth. Affected infants have symptoms of anaemia; some have other congenital anomalies, such as short stature or abnormal thumbs. Treatment is by oral steroids; monthly red blood cell transfusions are given to children who are steroid unresponsive and some may also be offered stem cell transplantation.

Transient erythroblastopenia of childhood (TEC) is usually triggered by viral infections and has the same haematological features as Diamond–Blackfan anaemia. The main differences between them is that, unlike Diamond–Blackfan anaemia, transient erythroblastopenia of childhood always recovers, usually within several weeks, there is no family history or RPS gene mutations and there are no congenital anomalies.

Increased red cell destruction (haemolytic anaemia)

Haemolytic anaemia is characterised by reduced red cell lifespan due to increased red cell destruction in the circulation (intravascular haemolysis) or liver or spleen (extravascular haemolysis). The lifespan of a normal red cell is 120 days and the bone marrow produces 173 000 million red cells per day. In haemolysis, red cell survival may be reduced to a few days but bone marrow production can increase about eight-fold, so haemolysis only leads to anaemia when the bone marrow is no longer able to compensate for the premature destruction of red cells.

In children, unlike neonates, immune haemolytic anaemias are uncommon. The main cause of haemolysis in children is *intrinsic* abnormalities of the red blood cells:

- Red cell membrane disorders (e.g. hereditary spherocytosis)
- Red cell enzyme disorders (e.g. glucose-6-phosphate dehydrogenase deficiency)
- Haemoglobinopathies (abnormal haemoglobins, e.g. β-thalassaemia major, sickle cell disease).

Haemolysis from increased red cell breakdown leads to:

- Anaemia
- Hepatomegaly and splenomegaly
- Increased blood levels of unconjugated bilirubin
- Excess urinary urobilinogen.

The diagnostic clues to haemolysis are:

- Raised reticulocyte count (on the blood film this is called 'polychromasia' as the reticulocytes have a characteristic lilac colour)
- Unconjugated bilirubinaemia and increased urinary urobilinogen
- Abnormal appearance of the red cells on a blood film (e.g. spherocytes, sickle shaped or very hypochromic) (Fig. 22.6)
- Positive direct antiglobulin test (only if an immune cause, as this test identifies antibody-coated red blood cells)
- Increased red blood cell precursors in the bone marrow.

Hereditary spherocytosis

HS occurs in 1 in 5000 births in Caucasians. It usually has an autosomal dominant inheritance, but in 25% there is no family history and it is caused by new mutations. The disease is caused by mutations in genes for proteins of the red cell membrane (mainly spectrin, ankyrin or band 3). This results in the red cell losing part of its membrane when it passes through the spleen. This reduction in its surface-to-volume ratio causes the cells to become spheroidal, making them less deformable than normal red blood cells and leads to their destruction in the microvasculature of the spleen.

Clinical features

The disorder is often suspected because of the family history. The clinical manifestations are highly variable. Although hereditary spherocytosis-affected individuals may be completely asymptomatic, the clinical features include:

- Jaundice – usually develops during childhood but may be intermittent; may cause severe haemolytic jaundice in the first few days of life
- Anaemia – presents in childhood with mild anaemia (haemoglobin 9–11 g/dl), but the haemoglobin level may transiently fall during infections
- Mild to moderate splenomegaly – depends on the rate of haemolysis
- Aplastic crisis – uncommon, transient (2–4 weeks), caused by parvovirus B19 infection
- Gallstones – due to increased bilirubin excretion.

Diagnosis

The blood film is usually diagnostic but more specific tests are available (e.g. osmotic fragility, dye binding tests), although seldom required. Autoimmune haemolytic anaemia is also associated with spherocytes and this should be excluded with a direct antibody test in the absence of a family history of hereditary spherocytosis.

Management

Most children have mild chronic haemolytic anaemia and the only treatment they require is oral folic acid as they have a raised folic acid requirement secondary to their increased red blood cell production. Splenectomy is beneficial but is only indicated for poor growth or troublesome symptoms of anaemia (e.g. severe tiredness, loss of vigour) and is usually deferred until after 7 years of age because of the risks of post-splenectomy sepsis. Prior to splenectomy all patients should be checked that they have been vaccinated against *Haemophilus influenzae* (Hib), meningitis C and *Streptococcus pneumoniae* and lifelong daily oral penicillin prophylaxis is advised. Aplastic crisis from parvovirus B19 infection usually requires one or two blood transfusions over 3–4 weeks when no red blood cells are produced. If gallstones are symptomatic, cholecystectomy may be necessary.

Glucose-6-phosphate dehydrogenase (G6PD) deficiency

G6PD deficiency is the commonest red cell enzymopathy affecting over 100 million people worldwide. It has a high prevalence (10–20%) in individuals originating from central Africa, the Mediterranean, the Middle East and the Far East. Many different mutations of the gene have been described, leading to different clinical features in different populations.

G6PD is the rate-limiting enzyme in the pentose phosphate pathway and is essential for preventing oxidative damage to red cells. Red cells lacking G6PD are susceptible to oxidant-induced haemolysis. G6PD deficiency is X-linked and therefore predominantly affects males. Females who are heterozygotes are usually clinically normal as they have about half the normal G6PD activity. Females may be affected either if they are homozygous or, more commonly, when by chance more of the normal than the abnormal X chromosomes have been inactivated (extreme Lyonisation – the Lyon hypothesis is that, in every XX cell, one of the X chromosomes is inactivated and that this is random). In Mediterranean, Middle Eastern and Oriental populations, affected males have very low or absent enzyme activity in their red cells. Affected Afro-Caribbeans have 10–15% normal enzyme activity.

Clinical manifestations

Children usually present clinically with:

- *Neonatal jaundice* – onset is usually in the first 3 days of life. Worldwide it is the most common cause of severe neonatal jaundice requiring exchange transfusion

Box 22.2 Drugs and chemicals which can cause haemolysis in children with G6PD deficiency

Antimalarials
- Primaquine
- Quinine
- Chloroquine

Antibiotics
- Sulphonamides (including co-trimoxazole)
- Quinolones (ciprofloxacin, nalidixic acid)
- Nitrofurantoin

Analgesics
- Aspirin (in high doses)

Chemicals
- Naphthalene (mothballs)
- Divicine (fava beans – also called broad beans)

Adapted from: *British National Formulary for Children 2009.*

- *Acute haemolysis* – precipitated by:
  - infection, the most common precipitating factor
  - certain drugs (see Box 22.2)
  - fava beans (broad beans)
  - naphthalene in mothballs.

Haemolysis due to G6PD deficiency is predominantly intravascular. This is associated with fever, malaise and the passage of dark urine, as it contains haemoglobin as well as urobilinogen. The haemoglobin level falls rapidly and may drop below 5 g/dl over 24–48 h.

Diagnosis

Between episodes, almost all patients have a completely normal blood picture and no jaundice or anaemia. The diagnosis is made by measuring G6PD activity in red blood cells. During a haemolytic crisis, G6PD levels may be misleadingly elevated due to the higher enzyme concentration in reticulocytes, which are produced in increased numbers in response to the destruction of mature red cells. A repeat assay is then required in the steady state to confirm the diagnosis.

Management

The parents should be given advice about the signs of acute haemolysis (jaundice, pallor and dark urine) and provided with a list of drugs, chemicals and food to avoid (Box 22.2). Transfusions are rarely required, even for acute episodes.

Haemoglobinopathies

These are red blood cell disorders which cause haemolytic anaemia because of reduced or absent production of HbA (α- and β-thalassaemias) or because of the production of an abnormal Hb (e.g. sickle cell disease). α-Thalassaemias are caused by deletions (occasionally mutations) in the α-globin gene. β-Thalassaemia and

Table 22.2 Haemoglobins in haemoglobinopathies

	HbA	HbA_2	HbF	HbS
Newborn	25%	1%	74%	–
Adult	97%	2%	–	–
β-Thalassaemia trait	>90%	↑	+ ↑	–
β-Thalassaemia major	–	↑	↑	–
Sickle cell trait	✓	✓	+ ↑	✓
Sickle cell disease	–	✓	+ ↑	✓

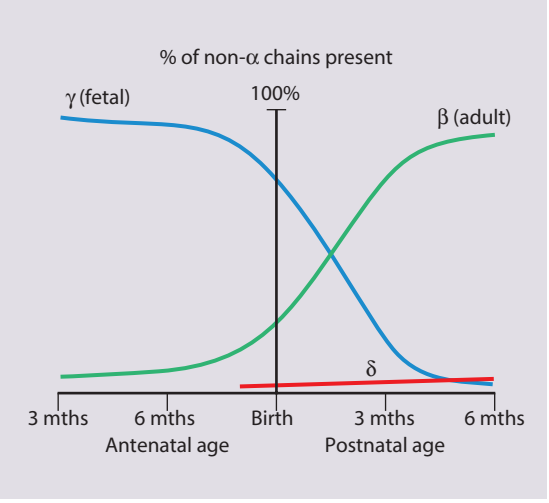


Figure 22.7 Changes in haemoglobin chains in the fetus and infancy.

sickle cell disease are caused by mutations in the β-globin gene. Clinical manifestations of the haemoglobinopathies affecting the β-chain are delayed until after 6 months of age when most of the HbF present at birth has been replaced by adult HbA (Fig. 22.7, Table 22.2).

Sickle cell disease

This is now the commonest genetic disorder in children in many European countries, including the UK (prevalence 1 in 2000 live births). Sickle cell disease is the collective name given to haemoglobinopathies in which HbS is inherited. HbS forms as a result of a point mutation in codon 6 of the β-globin gene, which causes a change in the amino acid encoded from glutamine to valine. Sickle cell disease is most common in patients whose parents are black and originate from tropical Africa or the Caribbean but it is also found in the Middle East and in low prevalence in most other parts of the world except for northern Europeans.

There are three main forms of sickle cell disease and the sickle trait:

- *Sickle cell anaemia* (*HbSS*) – patients are homozygous for HbS, i.e. virtually all their Hb is HbS; they have small amounts of HbF and *no* HbA because they have the sickle mutation in both β-globin genes.
- *HbSC disease* (*HbSC*) – affected children inherit HbS from one parent and HbC from the other parent (HbC is formed as a result of a different point mutation in β-globin), so they also have *no* HbA because they have no normal β-globin genes.
- *Sickle β-thalassaemia* – affected children inherit HbS from one parent and β-thalassaemia trait from the other. They have no normal β-globin genes and most patients can make no HbA and therefore have similar symptoms to those with sickle cell anaemia.
- *Sickle trait* – inheritance of HbS from one parent and a normal β-globin gene from the other parent, so approximately 40% of the haemoglobin is HbS. They do not have sickle cell disease but are carriers of HbS, so can transmit HbS to their offspring. They are asymptomatic and are only identified as a result of blood tests.

Pathogenesis

In all forms of sickle cell disease, HbS polymerises within red blood cells forming rigid tubular spiral bodies which deform the red cells into a sickle shape. Irreversibly sickled red cells have a reduced lifespan and may be trapped in the microcirculation, resulting in blood vessel occlusion (vaso-occlusion) and therefore ischaemia in an organ or bone. This is exacerbated by low oxygen tension, dehydration and cold.

The clinical manifestations of sickle cell disease vary widely between different individuals. Disease severity also varies with different forms of sickle cell disease; in general, HbSS is the most severe form of the disease. Some patients produce more HbF (e.g. 10–15% of their Hb may be HbF, while most patients with sickle cell disease have HbF levels of 1%) and this results in a marked reduction in disease severity.

Clinical features

These are listed in Figure 22.8.

Management

Prophylaxis – Because of increased susceptibility to infection, especially encapsulated organisms, e.g. *Streptococcus pneumoniae* and *Haemophilus influenzae* type B because of functional asplenia, children should be fully immunised, including against pneumococcal, *Haemophilus influenzae* type B and meningococcus infection. To ensure full coverage of all pneumococcal subgroups, daily oral penicillin throughout childhood should be given. Patients should receive once-daily oral folic acid because of the increased demand for folic acid caused by the chronic haemolytic anaemia. Vaso-occlusive crises should be minimised by avoiding exposure to cold, dehydration, excessive exercise, undue stress or hypoxia. This requires practical measures such as dressing children warmly, giving drinks especially before exercise and taking extra care to keep children warm after swimming or when playing outside in the winter.

Clinical manifestations of sickle cell disease

Anaemia	All have moderate anaemia (usually Hb 6–10 g/dl) with clinically detectable jaundice from chronic haemolysis
Infection	All have marked increase in susceptibility to infection from encapsulated organisms such as pneumococci and *Haemophilus influenzae*. There is also an increased incidence of osteomyelitis caused by *Salmonella* and other organisms. This susceptibility to infection is due to **hyposplenism** secondary to chronic sickling and **microinfarction** in the spleen in infancy. The risk of overwhelming sepsis is greatest in early childhood
Painful crises	**Vaso-occlusive crises** causing pain affect many organs of the body with varying frequency and severity. A common mode of presentation in late infancy is the hand-foot syndrome, in which there is dactylitis with swelling and pain of the fingers and/or feet from vaso-occlusion (Fig. 22.9). The bones of the limbs and spine are the most common sites. The most serious type of painful crisis is actute chest syndrome, which can lead to severe hypoxia and the need for mechanical ventilation and emergency transfusion. Avascular necrosis of the femoral heads may also occur. Acute vaso-occlusive crises may be precipitated by exposure to cold, dehydration, excessive exercise or stress, hypoxia or infection
Acute anaemia	Sudden drop in haemoglobin from: **Haemolytic crises** – sometimes associated with infection **Aplastic crises** – haemoglobin may fall precipitously. Parvovirus infection causes complete, though temporary, cessation of red blood cell production **Sequestration crises** – sudden splenic or hepatic enlargement, abdominal pain and circulatory collapse from accumulation of sickled cells in spleen
Priapism	Needs to be treated promptly with exchange transfusion as it may lead to fibrosis of the corpora cavernosa and subsequent erectile impotence
Splenomegaly	Common in young children, but becomes much less frequent in older children
Long-term problems	**Short stature and delayed puberty** **Stroke and cognitive problems** – although 1 in 10 children with sickle cell disease have a stroke, twice that number develop more subtle neurological damage (Fig. 22.10), often manifest with poor concentration and school performance **Adenotonsillar hypertrophy** – causing sleep apnoea syndrome leading to nocturnal hypoxaemia, which can cause vaso-occlusive crises and/or stroke **Cardiac enlargement** – from chronic anaemia **Heart failure** – from uncorrected anaemia **Renal dysfunction** – may exacerbate enuresis because of inability to concentrate urine **Pigment gallstones** – due to increased bile pigment production **Leg ulcers** – uncommon in children **Psychosocial problems** – difficulties with education and behaviour exacerbated by time off school may occur

Figure 22.8 Clinical manifestations of sickle cell disease.



Figure 22.9 Dactylitis in sickle cell disease.

Treatment of acute crises – Painful crises should be treated with oral or intravenous analgesia according to need (may require opiates) and good hydration (oral or intravenous as required); infection should be treated with antibiotics; oxygen should be given if the oxygen saturation is reduced. Exchange transfusion is indicated for acute chest syndrome, stroke and priapism.

Treatment of chronic problems – Children who have recurrent hospital admissions for painful vaso-occlusive crises or acute chest syndrome (see Case History 22.2)

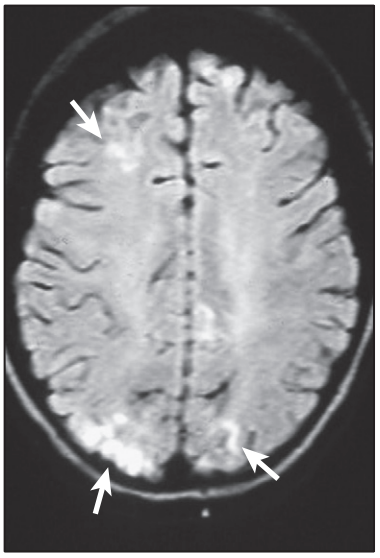


Figure 22.10 MRI of the brain in sickle cell disease showing multiple cerebral infarcts.

may benefit from hydroxyurea, a drug which increases their HbF production and helps protect against further crises. It requires monitoring for side-effects, especially white blood cell suppression. The most severely affected children (1–5%) who have had a stroke or who do not respond to hydroxyurea may be offered a bone marrow transplant. This is the only cure for sickle cell disease but can only be safely carried out if the child has an HLA-identical sibling who can donate their bone marrow – the cure rate is 90% but there is a 5% risk of fatal transplant-related complications.

Prognosis

Sickle cell disease is a cause of premature death due to one or more of these severe complications; around 50% of patients with the most severe form of sickle cell disease die before the age of 40 years. However, the mortality rate during childhood is around 3%, usually from bacterial infection.

Prenatal diagnosis and screening

Many countries with a high prevalence of haemoglobinopathies, including the UK, perform neonatal screening on dried blood spots (Guthrie test) collected in the first week of life. Early diagnosis of sickle cell disease allows penicillin prophylaxis to be started in early infancy instead of awaiting clinical presentation, possibly due to a severe infection. Prenatal diagnosis can be carried out by chorionic villus sampling at the end of the first trimester if parents wish to choose this option to prevent the birth of an affected child.

SC disease

Children with SC disease usually have a nearly normal haemoglobin level and fewer painful crises than those with HbSS, but they may develop proliferative retinopathy in adolescence. Their eyes should be checked periodically. They are also prone to develop osteonecrosis of the hips and shoulders.

Case History

22.2 Acute sickle chest syndrome

Princess, a 9-year-old girl with known sickle cell anaemia (HbSS), presented with increasing chest pain for 6 h. She had a non-productive cough. On examination, she had a fever of 39.7°C. Her breathing was laboured, respiratory rate increased and there was reduced air entry at both bases.

Investigations

- Haemoglobin 6 g/dl, WBC 14×10^9/L, platelets 350×10^9/L
- Chest X-ray (see Fig. 22.11)
- Oxygen saturation – 89% in air
- Arterial PO_2 – 9.3 kPa (70 mmHg) breathing face-mask oxygen
- Blood cultures were taken and viral titres performed.

A diagnosis of acute sickle chest syndrome was made, a potentially fatal condition. She was given oxygen by CPAP (continuous positive airways pressure). An exchange transfusion was performed. Broad-spectrum antibiotics were commenced. She responded well to treatment.

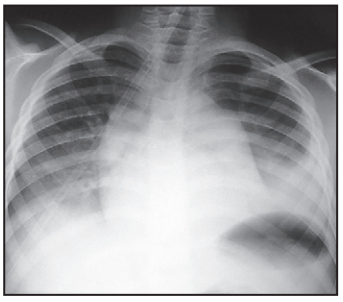


Figure 22.11 Chest X-ray in acute sickle chest syndrome showing bilateral lower zone consolidation. (Courtesy of Dr Parviz Habibi.)

Sickle cell trait (AS)

This is asymptomatic and rarely causes problems except under conditions of low oxygen tension. General anaesthesia does not constitute a risk in this population as long as they have been identified and hypoxia avoided.

β-Thalassaemias

The β-thalassaemias occur most often in people from the Indian subcontinent, Mediterranean and Middle East (Fig. 22.12). In the UK, most affected children are born to parents from the Indian subcontinent; in the past, many were born to Greek Cypriots, but this has become uncommon through active genetic counselling within their community.

There are two main types of β thalassaemia – both of which are characterised by a severe reduction in the production of β-globin (and thereby reduction in HbA production). All affected individuals have a

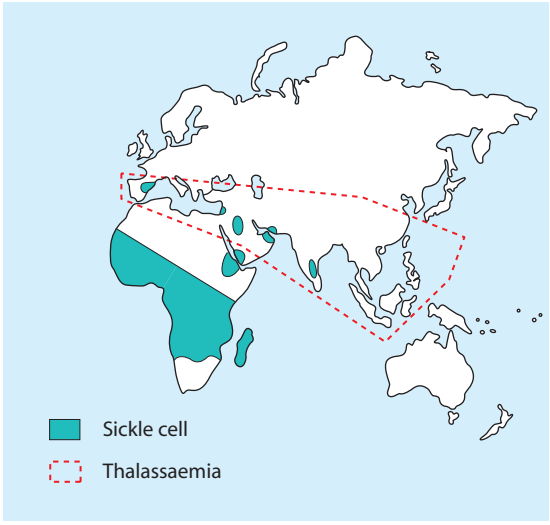


Figure 22.12 Ethnic origin of most families with sickle cell disease and thalassaemia.

severe reduction in β-globin and disease severity depends on the amount of residual HbA and HbF production.

- *β-Thalassaemia major* – This is the most severe form of the disease. HbA ($\alpha_2\beta_2$) cannot be produced because of the abnormal β-globin gene.
- *β-Thalassaemia intermedia* – This form of the disease is milder and of variable severity. The β-globin mutations allow a small amount of HbA and/or a large amount of HbF to be produced.

Clinical features (Fig. 22.13)

These are:

- Severe anaemia, which is transfusion dependent, from 3–6 months of age and jaundice
- Failure to thrive/growth failure
- Extramedullary haemopoiesis, prevented by regular blood transfusions. In the absence of regular blood transfusion, develop hepatosplenomegaly and bone marrow expansion; the latter leads to the classical facies with maxillary overgrowth and skull bossing (very rare in the UK and developed countries).

Management

The condition is uniformly fatal without regular blood transfusions, so all patients are given lifelong monthly transfusions of red blood cells. The aim is to maintain the haemoglobin concentration above 10 g/dl in order to reduce growth failure and prevent bone deformation. Repeated blood transfusion causes chronic iron overload, which causes cardiac failure, liver cirrhosis, diabetes, infertility and growth failure. For this reason, all patients are treated with iron chelation with subcutaneous desferrioxamine, or with an oral iron chelator drug, such as deferasirox, starting from 2 to 3 years of age. Patients who comply well with transfusion and chelation have a 90% chance of living into their forties and beyond. However, compliance is difficult. Those who cannot comply have a high mortality in early adulthood from iron overload. The complications of multiple transfusions are shown in Box 22.3. An alternative treatment for β-thalassaemia major is bone marrow transplantation, which is currently the only cure. It is generally reserved for children with an HLA-identical sibling as there is then a 90–95% chance of success (i.e. transfusion independence and long-term cure) but a 5% chance of transplant-related mortality.

Prenatal diagnosis

For parents who are both heterozygous for β-thalassaemia, there is a 1 in 4 risk of having an affected child. Prenatal diagnosis of β-thalassaemia (DNA analysis of a chorionic villus sample) should be offered together with genetic counselling to help parents to make informed decisions about whether or not to continue the pregnancy.

β-Thalassaemia trait

Heterozygotes are usually asymptomatic. The red cells are hypochromic and microcytic. Anaemia is mild or absent, with a disproportionate reduction in MCH (18–22 fl) and MCV (60–70 fl). The red blood cell count is therefore usually increased (>5.5×10^{12}/L). The most important diagnostic feature is a raised HbA_2, usually about 5% , and in about half there is a mild elevation of HbF level of 1–3%. β-Thalassaemia trait can cause confusion with mild iron deficiency because of the hypochromic/microcytic red cells but can be distinguished by measuring serum ferritin, which is low in iron deficiency but not β-thalassaemia trait.

To avoid unnecessary iron therapy, serum ferritin levels should be measured in patients with mild anaemia and microcytosis prior to starting iron supplements.

α-Thalassaemias

Healthy individuals have four α-globin genes. The manifestation of α-thalassaemia syndromes depends on the number of functional α-globin genes.

The most severe α-thalassaemia, *α-thalassaemia major* (also known as Hb Barts hydrops fetalis) is caused by deletion of all four α-globin genes, so no HbA ($\alpha_2\beta_2$) can be produced. It occurs mainly in families of South-east Asian origin and presents in mid-trimester with fetal hydrops (oedema and ascites) from fetal anaemia, which is always fatal in utero or within hours of delivery. The only long-term survivors of α-thalassaemia major are those who have received monthly intrauterine transfusions until delivery followed by lifelong monthly transfusions after birth. The diagnosis is made by Hb electrophoresis or Hb HPLC (high-performance liquid chromatography), which shows only Hb Barts.

When only three of the α-globin genes are deleted (*HbH disease*), affected children have mild–moderate anaemia but occasional patients are transfusion-dependent.

Deletion of one or two α-globin genes (known as α-thalassaemia trait) is usually asymptomatic and anaemia is mild or absent. The red cells may be hypochromic and microcytic, which may cause confusion with iron deficiency.

Clinical features and complications of β-thalassaemia major

Pallor

Jaundice

Bossing of the skull
Maxillary overgrowth

Splenomegaly and hepatomegaly

Need for repeated blood transfusions
Complications shown in Box 22.3



Figure 22.13 Facies in β-thalassaemia showing maxillary overgrowth and skull bossing in a child who has not been adequately transfused. This is now very rare in the UK and developed countries.

Box 22.3 Complications of long-term blood transfusion in children

Iron deposition – the most important (all patients)

- Heart – cardiomyopathy
- Liver – cirrhosis
- Pancreas – diabetes
- Pituitary gland – delayed growth and sexual maturation
- Skin – hyperpigmentation

Antibody formation (10% of children)

- Allo-antibodies to transfused red cells in the patient make finding compatible blood very difficult

Infection – now uncommon (<10% of children)

- Hepatitis A, B, C
- HIV
- Malaria
- Prions (e.g. new variant CJD)

Venous access (common problem)

- Often traumatic in young children
- Central venous access device (e.g. Portacath) may be required; these predispose to infection.

Anaemia in the newborn

Reduced red blood cell production

There are two main but rare causes in the newborn and both cause red cell aplasia:

- Congenital infection with parvovirus B19
- Congenital red cell aplasia (Diamond–Blackfan anaemia).

In this situation, the Hb is low and the red blood cells look normal. The diagnostic clue is that the reticulocyte count is low and the bilirubin is normal.

Increased red cell destruction (haemolytic anaemia)

This occurs either because of an antibody destroying the red blood cells (i.e. an extrinsic cause) or because there is an intrinsic abnormality of the surface or intracellular contents of the red blood cell. The main causes of haemolytic anaemia in neonates are:

- Immune (e.g. haemolytic disease of the newborn)
- Red cell membrane disorders (e.g. hereditary spherocytosis)
- Red cell enzyme disorders (e.g. glucose-6-phosphate dehydrogenase deficiency)
- Abnormal haemoglobins (e.g. α-thalassaemia major).

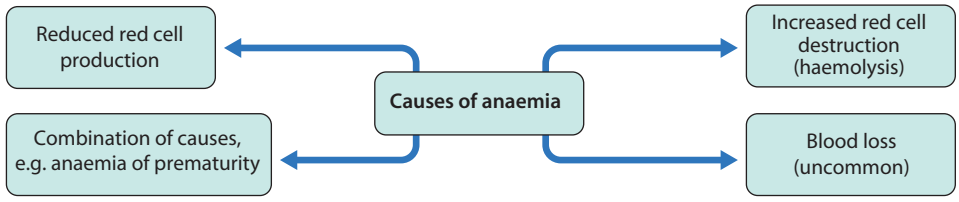
The diagnostic clues to a haemolytic anaemia are an increased reticulocyte count (due to increased red cell production to compensate for the anaemia) and increased unconjugated bilirubin (due to increased red cell destruction with release of this bile pigment into the plasma).

Haemolytic disease of the newborn (immune haemolytic anaemia of the newborn) is due to antibodies against blood group antigens. The most important are: anti-D (a 'rhesus' antigen), anti-A or anti-B (ABO blood group antigens) and anti-Kell. The mother is always negative for the relevant antigen (e.g. rhesus D-negative) and the baby is always positive; the mother then makes antibodies against the baby's blood group and these antibodies cross the placenta into the baby's circulation causing fetal or neonatal haemolytic anaemia. The diagnostic clue to this type of haemolytic anaemia is a positive direct anti-globulin test (Coombs test). This test is only positive in antibody-mediated anaemias and so is negative in all the other types of haemolytic anaemia. (These conditions are considered further in Chapter 10.)

The most common causes of non-immune haemolytic anaemia in neonates are: G6PD (glucose-6-

Summary

Anaemia



Reduced red cell production

Iron deficiency anaemia
- Common in infants and toddlers, especially if of Indian subcontinent origin
- Usually dietary in origin
- Occurs because of high iron requirement (1 mg/kg/day) for growth and body stores
- Will occur if infants are weaned at 6 months of age on to a mixed diet including iron-rich food
- Is diagnosed from a hypochromic microcytic anaemia and low serum ferritin
- Is treated with dietary advice and oral iron therapy for at least 3 months

Red cell aplasia
- Congenital red cell aplasia ('Diamond–Blackfan anaemia')
- Transient erythroblastopenia of childhood (TEC)
- Parvovirus B19 infection

Increased red cell production (haemolysis)

Hereditary spherocytosis
- Inheritance is autosomal dominant, but in 25% of cases there is no family history
- May cause early, severe jaundice in newborn infants
- Is often asymptomatic, but it may cause anaemia, jaundice, splenomegaly, aplastic crisis and gallstones
- Can usually be diagnosed from the blood film
- Treatment is with folic acid, splenectomy if symptomatic

β-Thalassaemia major
- Mutation of the β-globin gene results in an inability to produce HbA ($\alpha_2\beta_2$)
- Clinical features: severe anaemia, failure to thrive/growth failure and hepatosplenomegaly
- Condition is fatal without regular blood transfusions, but blood transfusions cause iron overload
- Iron chelation therapy with desferrioxamine or oral iron chelation is essential in all patients to minimise iron overload

β-Thalassaemia trait and α-thalassaemia trait
- Can cause diagnostic confusion with mild iron deficiency

α-Thalassaemia major
- Deletion of all 4 α-globin genes, α-thalassaemia major is fatal in utero (Hb Barts) or within hours of birth

Isoimmune
- Haemolytic disease of the newborn

Immune haemolytic anaemia

G6PD deficiency
- Affects over 100 million people worldwide, usually of Mediterranean, Middle East, Far East and Central African ethnicity
- Is X-linked and therefore predominantly affects males, but females may be affected
- May present with neonatal jaundice
- Causes acute intermittent haemolysis precipitated by infection, certain drugs, fava beans (broad beans) and naphthalene in mothballs
- Parents should be given a list of drugs, chemicals and food to avoid

Sickle cell disease
- Family usually originates from tropical Africa or the Caribbean
- Autosomal recessive
- Sickled red cells result in ischaemia in organs or bones
- Main clinical features are: anaemia, infection, painful crises, sequestration crises, splenomegaly in some young children, growth failure, gallstones, behaviour and learning problems
- The most serious clinical complications are bacterial infection, acute chest syndrome, strokes and priapism
- Management: prophylactic penicillin and immunisation; folic acid; maintain good hydration
  Treat crises: analgesia, hydration, antibiotics, exchange or blood transfusion as indicated
  Long-term: hydroxyurea or occasionally bone marrow transplant

phosphate dehydrogenase) deficiency and hereditary spherocytosis. Haemoglobinopathies, apart from α-thalassaemia, rarely present with clinical features in the neonatal period but are detected on neonatal haemoglobinopathy screening (Guthrie test).

Blood loss

The main causes are:

- Feto-maternal haemorrhage (occult bleeding into the mother)
- Twin-to-twin transfusion (bleeding from one twin into the other one)
- Blood loss around the time of delivery (e.g. placental abruption).

The main diagnostic clue is severe anaemia with a raised reticulocyte count and normal bilirubin.

Anaemia of prematurity

The main causes are:

- Inadequate erythropoietin production
- Reduced red cell lifespan
- Frequent blood sampling whilst in hospital
- Iron and folic acid deficiency (after 2–3 months).

Bone marrow failure syndromes

Bone marrow failure (also known as aplastic anaemia) is a rare condition characterised by a reduction or absence of all three main lineages in the bone marrow leading to peripheral blood pancytopenia. It may be inherited or acquired. The acquired cases may be due to viruses (especially hepatitis viruses), drugs (such as sulphonamides, chemotherapy) or toxins (such as benzene, glue); however, many cases are labelled as 'idiopathic' because a specific cause cannot be identified.

The condition may be partial or complete. It may start as failure of a single lineage but progress to involve all three cell lines.

The clinical presentation is with:

- Anaemia due to reduced red cell numbers
- Infection due to reduced white cell numbers (especially neutrophils)
- Bruising and bleeding due to thrombocytopenia.

Inherited aplastic anaemia

These disorders are all rare.

Fanconi anaemia

This is the most common inherited form of aplastic anaemia. It is an autosomal recessive condition. The majority of children have congenital anomalies, including short stature, abnormal radii and thumbs, renal malformations, microphthalmia and pigmented skin lesions. Children may present with one or more of these anomalies or with signs of bone marrow failure which do not usually become apparent until the age of 5 or 6 years. Neonates with Fanconi anaemia nearly always have a normal blood count but it can be diagnosed by demonstrating increased chromosomal breakage of peripheral blood lymphocytes. This test can be used to identify affected family members or for prenatal diagnosis. Affected children are at high risk of death from bone marrow failure or transformation to acute leukaemia. The recommended treatment is bone marrow transplantation using normal donor marrow from an unaffected sibling or matched unrelated marrow donor.

Shwachman–Diamond syndrome

This rare autosomal recessive disorder is characterised by bone marrow failure, together with signs of pancreatic exocrine failure and skeletal abnormalities. Most are caused by mutations in the SBDS gene, which can be used for identifying unusual cases or prenatal diagnosis. The most common haematological problem is an isolated neutropenia or mild pancytopenia. Like Fanconi anaemia, there is an increased risk of transforming to acute leukaemia.

Bleeding disorders

Normal haemostasis

Haemostasis describes the normal process of blood clotting. It takes place via a series of tightly regulated interactions involving cellular and plasma factors.

There are five main components:

1. *Coagulation factors* – are produced (mainly by the liver) in an inactive form and are activated when coagulation is initiated (usually by tissue factor (TF), which is released by vessel injury; see Fig. 22.14)
2. *Coagulation inhibitors* – these either circulate in plasma or are bound to endothelium and are necessary to prevent widespread coagulation throughout the body once coagulation has been initiated
3. *Fibrinolysis* – this process limits fibrin deposition at the site of injury due to activity of the key enzyme plasmin
4. *Platelets* – are vital for haemostasis as they aggregate at sites of vessel injury to form the primary haemostatic plug which is then stabilised by fibrin
5. *Blood vessels* – both initiate and limit coagulation. Intact vascular endothelium secretes prostaglandin I_2 and nitric oxide (which promote vasodilatation and inhibit platelet aggregation). Damaged endothelium releases TF and procoagulants (e.g. collagen and von Willebrand factor) and there are inhibitors of coagulation on the endothelial surface (thrombomodulin, antithrombin and protein S) to modulate coagulation.

The endpoint of the coagulation cascade is generation of thrombin. A simplified model is shown in Figure 22.14. The two main pathways for thrombin generation were identified many years ago as the intrinsic and extrinsic pathways. Important components of these pathways are still being discovered. In recent years, the crucial role of tissue factor (TF) in haemostasis has been recognised and it is now thought that the extrinsic pathway is the one primarily responsible for initiating both normal haemostasis and thrombotic disease.

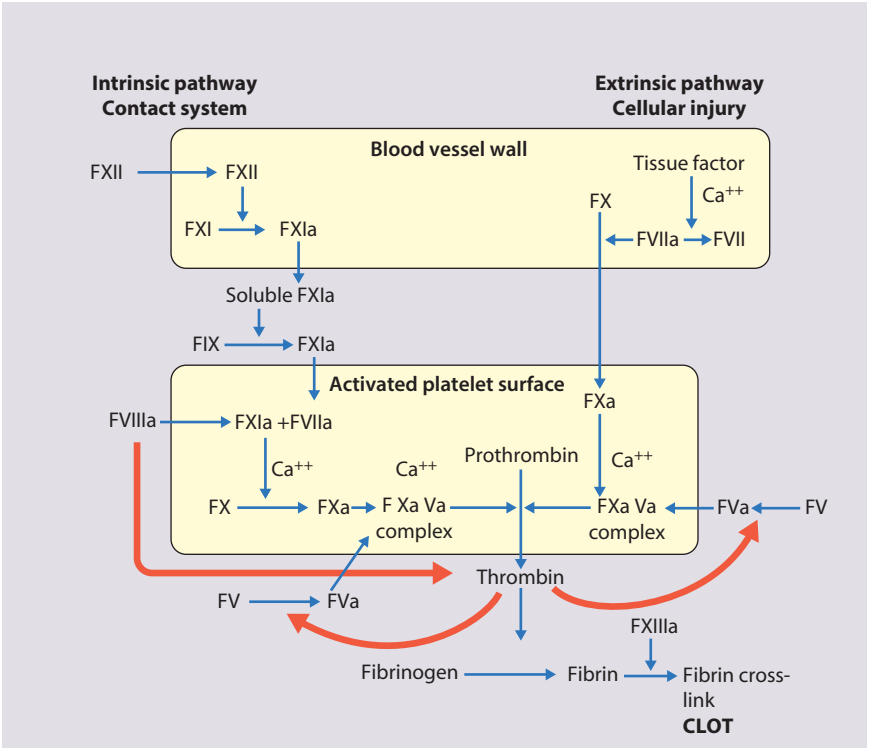


Figure 22.14 Schematic representation of the coagulation pathway.

Box 22.4 Helpful clinical features in evaluating bleeding disorders

Age of onset

- Neonate – in 20% of haemophilias, bleeding occurs in the neonatal period, usually with intracranial haemorrhage or bleeding after circumcision
- Toddler – haemophilias may present when starting to walk
- Adolescent – von Willebrand disease may present with menorrhagia

Family history

- Family tree – detailed family tree required
- Gender of affected relatives (if all boys, suggests haemophilia)

Bleeding history

- Previous surgical procedures and dental extractions – if uncomplicated, suggests bleeding tendency is acquired rather then inherited
- Presence of systemic disorders
- Drug history, e.g. anticoagulants
- Unusual pattern or inconsistent history – consider non-accidental injury

Pattern of bleeding

- Mucous membrane bleeding and skin haemorrhage – characteristic of platelet disorders or von Willebrand disease
- Bleeding into muscles or into joints – characteristic of haemophilia
- Scarring and delayed haemorrhage – suggestive of disorders of connective tissue, e.g. Marfan syndrome, osteogenesis imperfecta or factor XIII deficiency.

Diagnostic approach

Defects in the coagulation factors, in platelet number or function or in the fibrinolytic pathway are associated with an increased risk of bleeding. In contrast, defects in the naturally occurring inhibitors of coagulation (e.g. antithrombin) or in the vessel wall (e.g. damage from vascular catheters) are associated with thrombosis. In some cases, both pro- and anticoagulant abnormalities can occur at the same time, as seen in disseminated intravascular coagulation (DIC).

The diagnostic evaluation of an infant or child for a possible bleeding disorder includes:

- Identifying features in the clinical presentation that suggest the underlying diagnosis, as indicated in Box 22.4
- Initial laboratory screening tests to determine the most likely diagnosis (Table 22.3)

Table 22.3 Investigations in haemophilia A and von Willebrand disease

	Haemophilia A	von Willebrand disease
PT	Normal	Normal
APTT	↑↑	↑ or normal
Factor VIII:C	↓↓	↓ or normal
vWF Antigen	Normal	↓
RiCoF (activity)	Normal	↓
Ristocetin-induced platelet aggregation	Normal	Abnormal
vWF multimers	Normal	Variable

PT, prothrombin time; APTT, activated partial thromboplastin time; RiCoF, ristocetin co-factor, measures vWD activity.

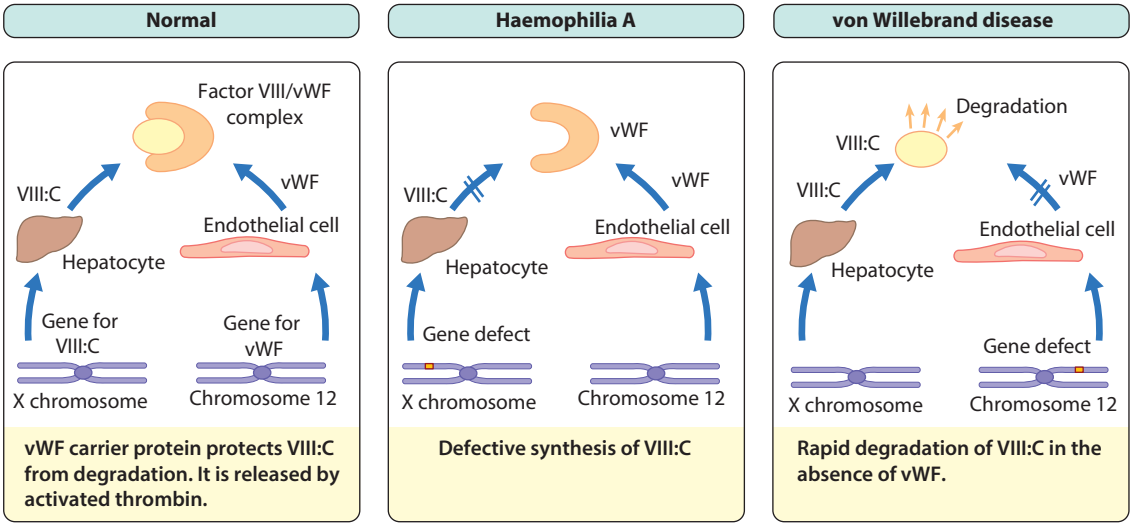


Figure 22.15 Factor VIII synthesis: normal, haemophilia A and von Willebrand disease.

- Specialist investigation to characterise a deficiency or exclude important conditions that can present with normal initial investigations, e.g. mild von Willebrand disease, factor XIII deficiency and platelet function disorders.

The most useful initial screening tests are:

- Full blood count and blood film
- Prothrombin time (PT) – measures the activity of factors II, V, VII and X
- Activated partial thromboplastin time (APTT) – measures the activity of factors II, V, VIII, IX, X, XI and XII
- If PT or APTT is prolonged, a 50:50 mix with normal plasma will distinguish between possible factor deficiency or presence of inhibitor
- Thrombin time – tests for deficiency or dysfunction of fibrinogen
- Quantitative fibrinogen assay
- D-dimers – to test for fibrin degradation products
- Biochemical screen, including renal and liver function tests.

The 'bleeding time' is no longer used to investigate platelet disorders, as it is unreliable. It has been replaced by in vitro tests of platelet function on a platelet function analyser, which can be performed on a peripheral blood sample.

In the neonate, the levels of all clotting factors except factor VIII (FVIII) and fibrinogen are lower; preterm infants have even lower levels. Therefore the results have to be compared with normal values in infants of a similar gestational and postnatal age. In view of this, and since it is often difficult to obtain good-quality neonatal samples, it is sometimes necessary to exclude an inherited coagulation factor deficiency by testing the coagulation of both parents.

Haemophilia

The commonest severe inherited coagulation disorders are haemophilia A and haemophilia B. Both have X-linked recessive inheritance. In haemophilia A, there is FVIII deficiency (Fig. 22.15); it has a frequency of 1 in 5000 male births. Haemophilia B (FIX deficiency)

Table 22.4 Severity of haemophilia

Factor VIII:C	Severity	Bleeding tendency
<1%	Severe	Spontaneous joint/ muscle bleeds
1–5%	Moderate	Bleed after minor trauma
>5–40%	Mild	Bleed after surgery



Figure 22.16 Severe arthropathy from recurrent joint bleeds in haemophilia. The aim of modern management is to prevent this from occurring.

has a frequency of 1 in 30 000 male births. Two-thirds of newly diagnosed infants have a family history of haemophilia, whereas one-third are sporadic. Identifying female carriers requires a detailed family history, analysis of coagulation factors and DNA analysis. Prenatal diagnosis is available using DNA analysis.

Clinical features

The disorder is graded as severe, moderate or mild, depending on the FVIII:C (or IX:C in haemophilia B) level (Table 22.4). The hallmark of severe disease is recurrent spontaneous bleeding into joints and muscles, which can lead to crippling arthritis if not properly treated (Fig. 22.16). Most children present towards the end of the first year of life, when they start to crawl or walk (and fall over). Bleeding episodes are most frequent in joints and muscles. Where there is no family history, non-accidental injury may initially be suspected. Almost 40% of cases present in the neonatal period, particularly with intracranial haemorrhage, bleeding post-circumcision or prolonged oozing from heel stick and venepuncture sites. The severity usually remains constant within a family.

Management

Recombinant FVIII concentrate for haemophilia A or recombinant FIX concentrate for haemophilia B is given by prompt intravenous infusion whenever there is any bleeding. If recombinant products are unavailable, highly purified, virally inactivated plasma-derived products should be used. The quantity required depends on the site and nature of the bleed. In general, raising the circulating level to 30% of normal is sufficient to treat minor bleeds and simple joint bleeds. Major surgery or life-threatening bleeds require the level to be raised to 100% and then maintained at 30–50% for up to 2 weeks to prevent secondary haemorrhage. This can only be achieved by regular infusion of factor concentrate (usually 8–12-hourly for FVIII, 12–24-hourly for FIX, or by continuous infusion) and by closely monitoring plasma levels. *Intramuscular injections, aspirin and non-steroidal anti-inflammatory drugs should be avoided in all patients with haemophilia.*

Complications are listed in Box 22.5.

Home treatment is encouraged to avoid delay in treatment, which increases the risk of permanent damage, e.g. progressive arthropathy. Parents are usually taught to give replacement therapy at home when the child is 2–3 years of age and many children are able to administer their own treatment from 7–8 years of age.

Prophylactic FVIII is given to all children with severe haemophilia A to further reduce the risk of chronic joint damage by raising the baseline level above 2%. Primary prophylaxis usually begins at age 2–3 years, and is given two to three times per week. If peripheral venous access is poor, a central venous access device (e.g. Portacath) may be required. Prophylaxis has been shown to result in better joint function in adult life. Similarly, patients with severe haemophilia B are usually given prophylactic FIX.

Desmopressin (DDAVP) may allow mild haemophilia A to be managed without the use of blood products. It is given by infusion and stimulates endogenous release of FVIII:C and von Willebrand factor (vWF). Adequate levels can be achieved to enable minor surgery and dental extraction to be undertaken. DDAVP is ineffective in haemophilia B.

Haemophilia centres should supervise the management of children with bleeding disorders. They provide a multidisciplinary approach with expert medical,

Box 22.5 Complications of treatment of haemophilia

Inhibitors, i.e. antibodies to FVIII or FIX

- Develop in 5–20%
- Reduce or completely inhibit the effect of treatment
- Require the use of very high doses of factor VIII or bypassing agents (e.g. FVIIa) for treating bleeding
- May be amenable to immune tolerance induction

Transfusion-transmitted infections

- Hepatitis A, B and C
- HIV
- ?Prions

Vascular access

- Peripheral veins – may be difficult to cannulate
- Central venous access devices may become infected or thrombosed.

nursing and laboratory input. Specialised physiotherapy is needed to preserve muscle strength and avoid damage from immobilisation. Psychosocial support is an integral part of maintaining compliance.

Self-help groups such as the Haemophilia Society may provide families with helpful information and support.

von Willebrand disease (vWD)

Von Willebrand factor (vWF) has two major roles:

- It facilitates platelet adhesion to damaged endothelium
- It acts as the carrier protein for FVIII:C, protecting it from inactivation and clearance.

Von Willebrand disease (vWD) results from either a quantitative or qualitative deficiency of von Willebrand factor (vWF). This causes defective platelet plug formation and, since vWF is a carrier protein for FVIII:C, patients with vWD also are deficient in FVIII:C (see Fig. 22.15).

There are many different mutations in the vWF gene and many different types of vWD. The inheritance is usually autosomal dominant. The commonest subtype, type 1 (60–80%), is usually fairly mild and is often not diagnosed until puberty or adulthood.

Clinical features

These are:

- Bruising
- Excessive, prolonged bleeding after surgery
- Mucosal bleeding such as epistaxis and menorrhagia.

In contrast to haemophilia, spontaneous soft tissue bleeding such as large haematomas and haemarthroses are uncommon.

Management

Treatment depends on the type and severity of the disorder. Type 1 vWD can usually be treated with DDAVP, which causes secretion of both FVIII and vWF into plasma. DDAVP should be used with caution in children <1 year of age as it can cause hyponatraemia due to water retention and may cause seizures, particularly after repeated doses, and if fluid intake is not strictly regulated. More severe types of vWD have to be treated with *plasma-derived* FVIII concentrate, as DDAVP is ineffective and recombinant FVIII concentrate contains no vWF. Cryoprecipitate is no longer used to treat vWD as it has not undergone viral inactivation. *Intramuscular injections, aspirin and non-steroidal anti-inflammatory drugs should be avoided in all patients with vWD.*

Summary

The child with abnormal bleeding
– into soft tissues, mucocutaneous or following surgery

Acquired disorders

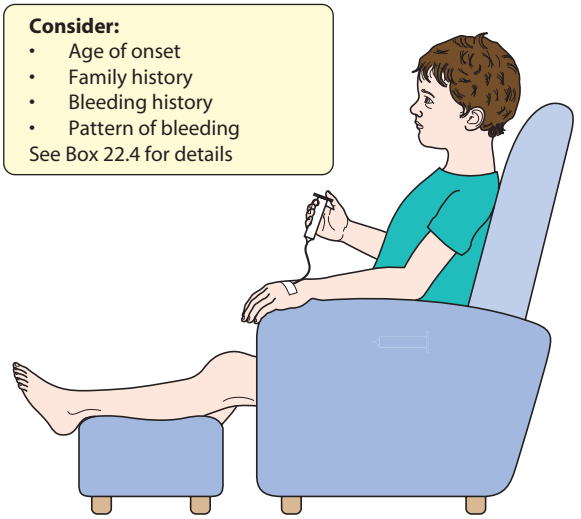
Vitamin K deficiency:
- mainly neonates or early infancy

Liver disease

Thrombocytopenia:
- immune, DIC, etc.

Consider:
- Age of onset
- Family history
- Bleeding history
- Pattern of bleeding

See Box 22.4 for details



Inherited disorders

Haemophilia A (factor VIII deficiency) and haemophilia B (factor IX deficiency):
- Are X-linked recessive disorders affecting males
- Presentation of severe disease – usually with recurrent spontaneous bleeding into joints and muscles at about 1 year of age
- Treatment – recombinant FVIII concentrate for haemophilia A or recombinant FIX concentrate for haemophilia B. Desmopressin (DDAVP) to treat mild haemophilia A
- Treatment complications – inhibitors and intravenous access

von Willebrand disease (vWD):
- Results from either a quantitative or qualitative deficiency of von Willebrand factor (vWF)
- Autosomal dominant
- Presentation – mucosal bleeding, e.g. epistaxis or menorrhagia in adolescence or excessive, prolonged bleeding after surgery
- Treatment – mild disease with DDAVP, severe disease with plasma-derived FVIII concentrate

Acquired disorders of coagulation

The main acquired disorders of coagulation affecting children are those secondary to:

- Haemorrhagic disease of the newborn due to vitamin K deficiency (see Ch. 10)
- Liver disease
- ITP (immune thrombocytopenia)
- DIC (disseminated intravascular coagulation).

Vitamin K is essential for the production of active forms of factors II, VII, IX, X and for the production of naturally occurring anticoagulants such as proteins C and S. Vitamin K deficiency therefore causes reduced levels of all of these factors. The main clinical consequence of this is a prolonged prothrombin time and an increased risk of bleeding. Children may become deficient in vitamin K due to:

- Inadequate intake (e.g. neonates, long-term chronic illness with poor intake)
- Malabsorption (e.g. coeliac disease, cystic fibrosis, obstructive jaundice)
- Vitamin K antagonists (e.g. warfarin).

Thrombocytopenia

Thrombocytopenia is a platelet count $<150\times10^9$/L. The risk of bleeding depends on the level of the platelet count:

- Severe thrombocytopenia (platelets $<20\times10^9$/L) – risk of spontaneous bleeding
- Moderate thrombocytopenia (platelets 20–50×10^9/L) – at risk of excess bleeding during operations or trauma but low risk of spontaneous bleeding
- Mild thrombocytopenia (platelets 50–150×10^9/L) – low risk of bleeding unless there is a major operation or severe trauma.

Table 22.5 Causes of purpura or easy bruising

Platelet count reduced, i.e. thrombocytopenia	
Increased platelet destruction or consumption	
Immune	ITP (immune thrombocytopenia)
	SLE (systemic lupus erythematosus)
	Alloimmune neonatal thrombocytopenia
Non-immune	Haemolytic uraemic syndrome
	Thrombotic thrombocytopenic purpura
	DIC (disseminated intravascular coagulation)
	Congenital heart disease
	Giant haemangiomas (Kasabach–Merritt syndrome)
	Hypersplenism
Impaired platelet production	
Congenital	Fanconi anaemia
	Wiskott–Aldrich syndrome
	Bernard–Soulier syndrome
Acquired	Aplastic anaemia
	Marrow infiltration (e.g. leukaemia)
	Drug-induced
Platelet count normal	
Platelet dysfunction	
Congenital	Rare disorders, e.g. Glanzmann thromboasthenia
Acquired	Uraemia, cardiopulmonary bypass
Vascular disorders	
Congenital	Rare disorders, e.g. Ehlers–Danlos, Marfan syndrome, hereditary haemorrhagic telangiectasia
Acquired	Meningococcal and other severe infections
	Vasculitis, e.g. Henoch–Schönlein purpura, SLE
	Scurvy

Thrombocytopenia may result in bruising, petechiae, purpura and mucosal bleeding (e.g. epistaxis, bleeding from gums when brushing teeth). Major haemorrhage in the form of severe gastrointestinal haemorrhage, haematuria and intracranial bleeding is much less common. The causes of easy bruising and purpura are listed in Table 22.5. While purpura may signify thrombocytopenia, it also occurs with a normal platelet count from platelet dysfunction and vascular disorders.

Immune thrombocytopenia (ITP)

Immune thrombocytopenia is the commonest cause of thrombocytopenia in childhood. It has an incidence of around 4 per 100 000 children per year. It is usually caused by destruction of circulating platelets by antiplatelet IgG autoantibodies. The reduced platelet count may be accompanied by a compensatory increase of megakaryocytes in the bone marrow.

Clinical features

Most children present between the ages of 2 and 10 years, with onset often 1–2 weeks after a viral infection. In the majority of children, there is a short history of days or weeks. Affected children develop petechiae, purpura and/or superficial bruising (see Case History 22.3). It can cause epistaxis and other mucosal bleeding but profuse bleeding is uncommon, despite the fact that the platelet count often falls to $<10\times10^9$/L. Intracranial bleeding is a serious but rare complication, occurring in 0.1–0.5%, mainly in those with a long period of severe thrombocytopenia.

Diagnosis

ITP is a diagnosis of exclusion, so careful attention must be paid to the history, clinical features and blood film to ensure that another more sinister diagnosis is not missed. In the younger child, a congenital cause (such as Wiskott–Aldrich or Bernard–Soulier syndromes) should be considered. Any atypical clinical features, such as the presence of anaemia, neutropenia, hepatosplenomegaly or marked lymphadenopathy, should prompt a bone marrow examination to exclude acute leukaemia or aplastic anaemia. A bone marrow examination should also be performed if the child is going to be treated with steroids, since this treatment may temporarily mask the diagnosis of acute lymphoblastic leukaemia (ALL). Inadvertent steroid therapy in undiagnosed ALL mimicking ITP will compromise the long-term outcome of such patients. Systemic lupus erythematosus (SLE) should also be considered. However, if the clinical features are characteristic, with no abnormality in the blood other than a low platelet count and no intention to treat, there is no need to examine the bone marrow.

Management

In about 80% of children, the disease is acute, benign and self-limiting, usually remitting spontaneously within 6–8 weeks. Most children can be managed at home and do not require hospital admission. Treatment is controversial. Most children do not need any therapy even if their platelet count is $<10\times10^9$/L but treatment should be given if there is evidence of major

Case History

22.3 Immune thrombocytopenic purpura (ITP)

Sian, aged 5 years, developed bruising and a skin rash over 24 h. She had had an upper respiratory tract infection the previous week. On examination she appeared well but had a purpuric skin rash with some bruises on the trunk and legs (Fig. 22.17). There were three blood blisters on her tongue and buccal mucosa, but no fundal haemorrhages, lymphadenopathy or hepatosplenomegaly. Urine was normal on dipsticks testing. A full blood count showed Hb 11.5 g/dl with normal indices, WBC and differential normal, platelet count 17×10^9/L. The platelets on the blood film were large; the film was otherwise normal. A diagnosis of ITP was made and she was discharged home. Her parents were counselled and given emergency contact names and telephone numbers. They were also given literature on the condition and advised that she should avoid contact sports but should continue to attend school. Over the next 2 weeks she continued to develop bruising and purpura but was asymptomatic. By the third week, she had no new bruises, and her platelet count was 25×10^9/L; the blood count and film showed no new abnormalities. The following week, the platelet count was 74×10^9/L and a week later it was 200×10^9/L. She was discharged from follow-up.

In immune thrombocytopenic purpura, in spite of impressive cutaneous manifestations and extremely low platelet count, the outlook is good and most will remit quickly without any intervention.



Figure 22.17 Bruising and purpura from immune thrombocytopenic purpura.

bleeding (e.g. intracranial or gastrointestinal haemorrhage) or persistent minor bleeding that affects daily lives such as excessive epistaxis or menstrual bleeding. The treatment options include oral prednisolone, intravenous anti-D or intravenous immunoglobulin and all have significant side-effects. Platelet transfusions are reserved for life-threatening haemorrhage as they raise the platelet count only for a few hours. The parents need immediate 24-hour access to hospital treatment, and the child should avoid trauma, as far as possible, and contact sports while the platelet count is very low.

Chronic ITP

In 20% of children, the platelet count remains low 6 months after diagnosis; this is known as chronic ITP. In the majority of children, treatment is mainly supportive; drug treatment is only offered to children with chronic persistent bleeding that affects daily activities or impairs quality of life. Children with significant bleeding are rare and require specialist care. A variety of treatment modalities are available, including rituximab, a monoclonal antibody directed against B lymphocytes. Newer agents such as thrombopoietic growth factors have shown clinical response in adults and may be used in children with severe non-responsive disease. Splenectomy can be effective for this group but is mainly reserved for children who fail drug therapy as it significantly increases the risk of infections and patients require lifelong antibiotic prophylaxis. If ITP in a child becomes chronic, regular screening for SLE should be performed, as the thrombocytopenia may predate the development of autoantibodies.

Summary

The child with petechiae or purpura

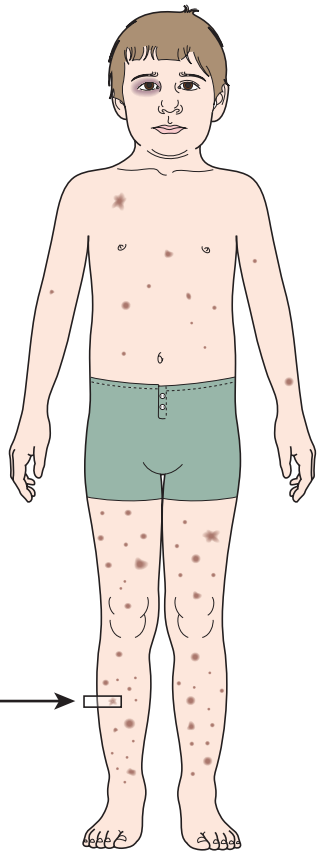
Non-thrombocytopenic

Henoch–Schönlein purpura
- Lesions confined to buttocks, extensor surfaces of legs and arms
- Swollen painful knees and ankles
- Abdominal pain
- Haematuria

Sepsis
- Meningococcal or viral
- Clinical features – fever, septicaemia, meningitis
- If suspected, give parenteral penicillin immediately

Trauma
- Accidental or non-accidental

Other causes (rare)



Positive glass test – rash does not blanch when pressed

Thrombocytopenia

Immune thrombocytopenia (ITP)
- 2–10 years
- Widespread petechiae and purpura and superficial bruising
- Distinguish from acute leukaemia and aplastic anaemia – clinical features, full blood count and blood film
- Bone marrow examination not required if only the platelet count is low, characteristic clinical features and no steroid treatment
- Is acute, benign and self-limiting in about 80% of children
- Treatment – controversial, usually not required unless there is bleeding

Leukaemia
- Clinical features – malaise, infection, pallor, hepatosplenomegaly, lymphadenopathy
- Blood count – also low Hb, blasts on film, confirmed on bone marrow

Disseminated intravascular coagulation (DIC)
- Critically ill – severe sepsis or shock or extensive tissue damage

Other causes (uncommon)

Disseminated intravascular coagulation

Disseminated intravascular coagulation (DIC) describes a disorder characterised by coagulation pathway activation leading to diffuse fibrin deposition in the microvasculature and consumption of coagulation factors and platelets.

The commonest causes of activation of coagulation are severe sepsis or shock due to circulatory collapse, e.g. in meningococcal septicaemia, or extensive tissue damage from trauma or burns. DIC may be acute or chronic and is likely to be initiated through the tissue factor pathway. The predominant clinical feature is bruising, purpura and haemorrhage. However, the pathophysiological process is characterised by microvascular thrombosis and purpura fulminans may occur.

No single test reliably diagnoses DIC. However, DIC should be suspected when the following abnormalities coexist – thrombocytopenia, prolonged prothrombin time (PT), prolonged APTT, low fibrinogen, raised fibrinogen degradation products and D-dimers and microangiopathic haemolytic anaemia. There is also usually a marked reduction in the naturally occurring anticoagulants, proteins C and S and antithrombin.

The most important aspect of management is to treat the underlying cause of the DIC (usually sepsis) while providing intensive care. Supportive care may be provided with fresh frozen plasma (to replace clotting factors), cryoprecipitate and platelets. Antithrombin and protein C concentrates have been used, particularly in severe meningococcal septicaemia with purpura fulminans. The use of heparin remains controversial.

Thrombosis in children

Thrombosis is uncommon in children and about 95% of venous thromboembolic events are secondary to underlying disorders associated with hypercoagulable states (see below). Thrombosis of cerebral vessels usually presents with signs of a stroke. (The condition is considered further in Chapters 10 and 27.) Rarely, children may inherit abnormalities in the coagulation and fibrinolytic pathway that increase their risk of developing clots even in the absence of underlying predisposing factors. These conditions are termed congenital prothrombotic disorders (thrombophilias). They are:

- Protein C deficiency
- Protein S deficiency
- Antithrombin deficiency
- Factor V Leiden
- Prothrombin gene G20210A mutation.

Proteins C and S and antithrombin are natural anticoagulants and their deficiencies are inherited in an autosomal dominant manner. Heterozygotes are also predisposed to thrombosis, usually venous, during the second or third decade of life and only rarely in childhood. Homozygous deficiency of protein C and protein S are very uncommon and present with life-threatening thrombosis with widespread haemorrhage and purpura into the skin (known as 'purpura fulminans') in the neonatal period. Homozygous antithrombin deficiency is not seen, probably because it is lethal in the fetus.

Factor V Leiden is an inherited abnormality in the structure of the coagulation protein factor V, which makes it resistant to degradation by activated protein C as part of the body's normal anticoagulant mechanism. The prothrombin gene mutation is associated with high levels of plasma prothrombin.

Acquired disorders are:

- Catheter-related thrombosis
- DIC (disseminated intravascular coagulation)
- Hypernatraemia
- Polycythaemia (e.g. due to congenital heart disease)
- Malignancy
- SLE (systemic lupus erythematosus) and persistent antiphospholipid antibody syndrome.

Diagnosis

Although inherited thrombophilia is very uncommon, these disorders predispose to life-threatening thrombosis and so it is important not to miss the diagnosis in any child presenting with an unexplained thrombotic event. Therefore, screening tests for the presence of an inherited thrombophilia should be carried out in the following situations:

- Any child with unanticipated or extensive venous thrombosis, ischaemic skin lesions or neonatal purpura fulminans
- Any child with a positive family history of neonatal purpura fulminans.

The screening tests are assays for proteins C and S, antithrombin assay, polymerase chain reaction (PCR) for factor V Leiden and for the prothrombin gene mutation.

Mutations in factor V (factor V Leiden) and the prothrombin gene, respectively, are present in 5% and 2% of the northern European population. Children with protein C deficiency or factor V Leiden have 4–6 times higher risk of developing recurrent thromboses. The risk increases significantly if these conditions are inherited together. Therefore it is reasonable to screen children who develop thrombosis for all of these factors in order to plan the best management to prevent thrombosis. In the UK, current practice is not to screen asymptomatic children for genetic defects, which are not going to affect their medical management, e.g. on the basis of family history alone, until they are old enough to receive appropriate counselling and make decisions for themselves.

Summary

Thrombosis

All children with thrombosis should be screened for inherited or acquired predisposing disorders.

Further reading

Bailey S, **Skinner R:** *Paediatric Haematology and Oncology. Oxford Specialist Handbooks in Paediatrics*. Oxford, 2009, Oxford University Press.

Lilleyman JS, **Hann IM**, **Blanchette VS, editors:** *Pediatric Haematology*, ed 3, Edinburgh, 2005, Churchill Livingstone.

A single-volume, comprehensive textbook

Orkin SH, **Fisher DE**, **Look T**, **et al., editors:** *Nathan and Oski's Hematology of Infancy and Childhood*, ed 7, Philadelphia, PA, 2008, Saunders.

Comprehensive, two-volume textbook.

Websites (Accessed May 2011)

British Committee for Standards in Haematology guidelines: Available at: http://www.bcshguidelines.com.

Diamond–Blackfan syndrome: UK Diamond Blackfan Anaemia Support Group. Available at: http://www.diamondblackfan.org.uk.

Haemophilia: World Federation of Haemophilia. Available at: http://www.wfh.org/index.asp?lang=EN or The Haemophilia Society. Available at: http://www.haemophilia.org.uk.

Shwachman–Diamond syndrome: Shwachman–Diamond Support UK. Available at: http://sdsuk.org.

Sickle Cell Society: Available at: http://www.sicklecellsociety.org.

UK Thalassaemia Society: Available at: http://www.ukts.org.

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Knowledge of children's emotions and behaviour is important in order to:

- know what constitutes the normal range
- understand common, normative and usually minor deviations and responses to stress, physical illness and injuries
- recognise and manage emotional and behavioural disorders.

Principles of normal development

Normal parenting

A child's behaviour, emotional responses and personality are the end result of interplay between genetic predisposition and environmental influences. The environment provides experience from which stems knowledge, learned behaviour or emotional responses and attitudes to oneself and the world. Interpersonal relationships are major environmental factors in promoting psychosocial development and the child's family is a principal source of these. Within the family, children should be protected, nurtured, educated and contained so that their development is supported optimally. Many societies are going through major demographic changes, including rising numbers of family breakdowns, lone-parent families and same gender parents. However, irrespective of the family structure, the essential elements of competent parenting or what is commonly referred to as 'good-enough parenting' are the same for all families (Box 23.1). Beyond this, the parents' attitude towards their children and how they handle their individual children help to determine how their personalities develop.

Normal early relationships

In the first 2 months of a baby's life, infants are not fussy about who responds to their needs. From 3 to 6 months they become more selective, demanding comfort from one or two caregivers. By 6–8 months they are particular about who responds to their needs or holds them, especially when distressed, and show tearful *separation anxiety* if their main caregiver, usually the mother, is not there. If tired, fearful, unhappy or in pain, they will cling to her and be comforted by her presence as an *attachment figure*. At this time, the child learns to crawl, and so is able to leave a primary caregiver and possibly encounter danger. The development of attachment behaviour allows the infant to keep track of their parent's whereabouts and resist separation. This close attachment relationship derives from social interaction and the mother's sensitive responsiveness to the baby's needs, not from any blood tie. It need not be with the biological mother, although it usually is. Its importance lies in it being:

- a particularly close relationship within which the child's development of trust, empathy, conscience and ideals is promoted, forming a prototype for future close relationships
- the child's primary source of comfort, providing the principal method of coping with stress (fear, anxiety, pain, etc.).

This underscores the importance of having a young child's parent 'rooming in' if admitted to hospital. Otherwise the child would be doubly distressed both by the absence of their attachment figure as well as by the threat of strange surroundings or procedures and by the stress of pain or illness.

If a young child is placed in strange, impersonal surroundings and separated from the mother for more than several hours, a *triphasic acute separation* reaction may set in (Fig. 23.1):

Box 23.1 Competent parents

- Are available physically and emotionally when needed
- Protect their children from harm
- Love their children – provide affection, support, comfort, food and shelter
- Exercise parental authority firmly, calmly and fairly, by setting clear and realistic limits for their children using rewards and praise to encourage compliance and non-physical sanctions to reduce non-compliance
- Respect their children's immature status and judge it accurately
- Keep adult business (sex, etc.) away from their children
- Model calm, positive and constructive problem-solving strategies for their children
- Have their own lives and do not live through their children
- Maintain their own self-esteem and personal development.

- Mounting anxiety about the fact that the child's mother fails to reappear produces distressed, irritable tearfulness (*protest*), which is hard to comfort
- After a day or two, this turns into a withdrawn state with no play, no interest in food and little speech or willingness for personal contact (*despair*)
- The child gradually cheers up from this but the close contact with the mother has been lost and the child is relatively indifferent to her when she reappears (*detachment*).

Recreating the original closeness can take weeks and is accompanied by a phase of irritability, misbehaviour and clinging. This can sometimes be seen when children who have been admitted to hospital as an emergency return home.

Children who have never had the opportunity for a close, secure attachment relationship in their early years are at risk of growing up as self-centred individuals who seek the affection and attention of others but have difficulty with close personal relationships and with learning to conform with social rules of conduct.

The selective clinging of early attachment behaviour diminishes over time so that in the second year of life children extend their emotional attachments to other family members and carers. By school age, they can tolerate separations from their parents for several hours. Children vary in their ability to do this depending on their temperament and social circumstances. For example, a child who is constitutionally apprehensive, who has an exceptionally anxious mother, or who has parents who threaten abandonment is likely to continue to cling to his/her mother for protection and comfort. A series of frightening events will tend to perpetuate clinging, which may persist well into middle childhood (age 5–12 years). This interferes with children's capacity to learn how to cope with anxiety on their own (Fig. 23.2).

The three stages of the acute separation response in children

Protest	Crying, distress Angry refusal to be comforted Asking for mummy
↓	
Despair	Moping Not playing Not eating
↓	
Detachment	Apparent cheering up and recovery Indifferent to parents on return to them

The above sequence develops over a period of days, but with considerable variation between children.

Figure 23.1 The three stages of the acute separation response in young children.



Figure 23.2 A 6-year-old with separation anxiety, showing what it feels like to leave her mother to go to school.

With entry into school, the importance of teachers and other children in shaping psychosocial development increases and their influence must be taken into account in understanding any schoolchild's development.

Summary

Early relationships:

Young children:

- develop a close attachment relationship with their mother (or main caregiver)
- if separated from their mother, may develop separation anxiety
- if admitted to hospital, should be able to have their parents stay with them.

Temperament

Children differ from each other in personality from birth, just as they do in physical appearance. This individuality in behavioural style or temperament is partly genetically determined. It is not fixed but changes slowly in the light of experience. It affects how other people deal with them.

A child born with a *difficult temperament* is prone to:

- predominantly negative moods – whinging, moaning, crying
- intense emotional reactions – screaming rather than whimpering, jumping for joy rather than smiling
- irregular biological functions – a lack of rhythm in sleeping, hunger or toileting
- negative initial responses to novel situations, e.g. pushing a new toy away
- protracted adjustment to new situations – taking weeks or months to settle into a new playgroup.

Such a pattern is a vulnerability factor for future emotional and behavioural problems. It may be hard for parents to maintain an affectionate relationship with a child who has a difficult temperament. Their self-confidence falters as they feel guilty that they have failed as parents. Such parents need support to maintain a positive, loving relationship with their child who will, if this can be done, soften and become easier to handle over a period of months. If parents lapse into irritable intolerance themselves, this is likely to maintain the child's grouchy and unsatisfied manner and may lead eventually to low self-esteem or the development of behavioural problems.

Self-esteem

Children develop views and make attributions about themselves. Most children experience praise and success in enough areas of their lives to develop a sense of inner self-confidence and self-worth. Those who do not are at increased risk of developing emotional and behavioural disorders which in turn may breed further shame and failure. A child who does not consider him/herself worthwhile and valued by others will play safe and not attempt new activities or explore new situations because of a fear of failure. This restricts the development of coping skills and knowledge of the world generally. It may also be a vulnerability factor for depression and anxiety disorders. Children who lack a belief in their own worth may adopt extraordinary and problematic behaviours in order to attract the attention and acclaim of others. For instance, one child took to openly eating dog faeces because it attracted a crowd of amazed children around her. Repeated failure, academically or socially, will undermine self-esteem, as will some disorders themselves (dyspraxia, enuresis and faecal soiling in particular). An important source of low self-esteem, however, is the child's parents, either because of their own low self-esteem or because of abuse (emotional, sexual or physical) and neglect.

Box 23.2 The quality of preschool thought

- The child is at the centre of his world ('I'm tired so it's getting dark')
- Everything has a purpose ('The sea is there for us to swim in')
- Inanimate objects are alive ('Naughty table hurt me') and have feelings and motives
- Poor categorisation (all men are Daddies)
- Use of magical thinking ('If I close my eyes, she'll go away')
- Use of sequences or routines rather than a sense of time
- The use of toys and other aspects of imaginative play as aids to thought (particularly in making sense of experience and social relationships).

Cognitive style

As children grow older, their thinking style evolves from one that is concrete to one that is able to cope with abstract thought. Below the age of about 5 years, thought is fundamentally egocentric, with the child being at the centre of his world (Box 23.2). During middle childhood, the dominant mode of thought is practical and orderly but tied to immediate circumstances and specific experiences rather than hypothetical possibilities or metaphors. Not until the mid-teens does the adult style of abstract thought begin to appear.

Adjust the way you talk to children to be compatible with their thinking style.

Coping with chronic or serious illness or adversities in childhood

Children can respond to adversity, including illness, in a number of ways:

- *Cognitive response* – can lie anywhere along the spectrum of over-acceptance to denial, with fluctuation over time. In over-acceptance, the child may allow the illness to overtake their life resulting in more impairment than is expected for level of symptoms, and high levels of anxiety about the slightest symptom. With denial, symptoms and warning signs may be ignored and treatment poorly adhered to.
- *Emotional response* – to diagnosis of illness and at times of relapse, may have similarities to a bereavement reaction or reaction to loss, with shock, denial, anger, followed by acceptance and adjustment (Fig. 23.3). Such responses to a serious illness are normal as long as the child proceeds through the phases.
- *Behavioural response* – young children tend to regress when stressed and behave younger than they actually are. A toddler may become

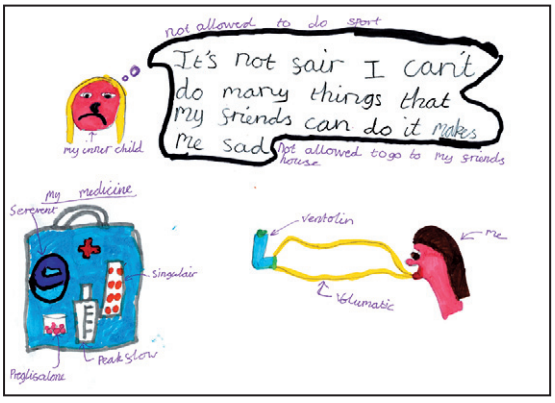


Figure 23.3 Drawing by a 10-year-old girl with asthma, showing how she felt about her illness. She was struggling to take her medicines and had multiple hospital admissions. Family therapy helped her to come to terms with her illness and take her medications.

overactive or clingy and display sleep and feeding difficulties. Regressive responses in older children predominantly manifest as problems with toileting, academic performance and peer relationships.

- *Somatic response* – can include expression of worry and distress through bodily symptoms such as recurrent abdominal pain.

Children suffering from chronic or serious illness are more vulnerable to mental health problems. This is related to:

- *Nature of illness* – this includes severity, chronicity, presence of constant discomfort and demands of treatment. Children with neurological disorders involving the brain, e.g. epilepsy, are at increased risk.
- *Stage of illness* – for example, diagnosis, deteriorations signalled by the need for new demanding treatments or by admissions to hospital.
- *Age of the child* – in infancy, illness may affect attachment and developmental milestones such as autonomy and mobility. Over 5 years of age there is a greater impact on educational progress, athletic activities and achievement. In adolescence, social adjustment, individual identity, independence from the family and poor adherence to treatment become more of an issue. Prolonged separation from parents resulting from illness will have a particularly negative impact between the age of 6 months and 3 years.
- *Temperament* – a child who is more adaptable to new situations will fare better. In contrast, a child who from early life has been difficult to soothe will fare worse
- *Intellectual capacity* – brighter children generally cope better.
- *Family factors* – the illness can have detrimental effects on the family and family difficulties can aggravate adaptation to illness.

Summary

Responses of children to illness or adversity include:

- Over-acceptance or refusal to accept the situation
- The sequence of shock, denial, anger, followed by acceptance and adjustment
- Regression of behaviour
- Somatic symptoms.

Adversities in the family

Family relationships are, for most children, the source of their most powerful emotions. Similarly, parents have more effect than anyone else on children's social learning and behaviour. The ecological model of child development indicates that families are generally the most potent environmental influence on a child's mental health. They are not all-powerful, since a predisposition to particular childhood emotional and behavioural problems can be inherited, but family influences interact with this so that overt disorder may or may not emerge. Not all disorders have their origin in family adversities: hyperkinetic disorder, tics and autism arise independently of them. Nevertheless, the non-genetic contribution of family interactions to emotional and behavioural disorders is often substantial and the mechanisms whereby they produce disorder are various. The following are some of the known risk factors:

- Angry discord between family members
- Parental mental ill health, especially maternal depression
- Divorce (Boxes 23.3 and 23.4) and bereavement
- Intrusive overprotection
- Lack of parental authority
- Physical and sexual abuse
- Emotional rejection or unremitting criticism
- Use of violence, terror, threats of abandonment or excessive guilt as disciplinary devices
- Taunting or belittlement of the child
- Inconsistent, unpredictable discipline
- Using the child to fulfil the unreasonable personal emotional needs of a parent
- Inappropriate responsibilities or expectations for the child's level of maturity.

Many of these risk factors can be aggravated by a difficult or unrewarding child whose behaviour or difficult temperament may make the adverse environment worse. While parents need to be made aware of changes required to improve the situation, it is unwise to blame them for causing their child's problem as it makes them less likely to engage in treatment.

Adversities may also arise outside the family. Experiences with other children are increasingly recognised as highly significant in psychosocial development. Bullying is a known adversity, as are instances of targeted

peer rejection, and other forms of peer-mediated persecution. With widespread access to the internet, cyber bullying is becoming an increasing problem. Merely being left out of things by other children (as opposed to being driven away) is much less pernicious. Conversely, having a number of steady, good-quality peer relationships is a marker for good prognosis in an emotional or behavioural problem which has resulted from environmental influences.

Summary

Regarding adversities

- The child's family is the most potent influence on the child's mental health
- Many but not all mental health problems originate from adversities in the family
- The child's temperament and adversities outside the family, e.g. bullying, may aggravate the situation.

Box 23.3 Reactions to parental divorce

Preschool
- Fear of further abandonment:
  - Intensified clinging at threatened separations
  - Sleep disturbances
  - Tearful, irritable and demanding
  - Disorganised play

Middle childhood
- Miserable at loss of parent, pining for restitution of family
- Self-blame in younger (age 6–8) age group
- Angry blaming of one parent for divorce in older (age 9–12) age group
- Loyalty conflicts
- Anxiety and jealousy at parents' new partners
- Educational underachievement

Adolescents
- Wide variation in response
- Various attempts to master the situation:
  - Detachment from family
  - Rapid maturation
  - Moral idealism
  - Critical of parents
- Educational underachievement
- Depression in some.

Box 23.4 Adjustment tasks facing children of divorced parents

- Acknowledgement of parental separation
- Regaining sense of direction in life activities
- Dealing with sense of loss and rejection
- Forgiving parents for break-up
- Accepting permanence of divorce, relinquishing wish for the previously intact family
- Feeling able to enter into new emotional relationships.

Problems of the preschool years

Meal refusal

A common scenario is a mother complaining that her child refuses to eat any or much of what she provides; mealtimes have become a battleground. Examination reveals a healthy, well-nourished child whose height and weight are securely within normal limits on a centile growth chart, or a small and thin child with faltering growth.

An account of what goes on at a typical mealtime may reveal:

- A past history of force-feeding
- Irregular meals so that the child is not predictably hungry
- Unsuitable meals
- Unreasonably large portions
- Multiple opportunities for distraction, e.g. TV.

Most importantly, how much does the child eat between meals? A well-nourished child is getting food from somewhere. Not all parents regard sweets and crisps as being food. Some mothers, while concerned about their child's apparently poor food intake, provide little variety in the child's diet. For strategies for dealing with meal refusal, see Box 23.5. Children with faltering growth may require specialist referral if they do not respond to this advice.

Sleep-related problems

Difficulty in settling to sleep at bedtime

This is a common problem in the toddler years. The child will not go to sleep unless the parent is present. Most instances are normal expressions of separation anxiety, but there may be other obvious reasons for it which can be explored in taking a history (Box 23.6), supplemented if necessary by the parents keeping a prospective sleep diary. Many cases will respond to simple advice:

- Creating a bedtime routine which cues the child to what is required
- Telling the child to lie quietly in bed until he/she falls asleep, recognising that children cannot fall asleep to order (although that is what everyone tells them to do).

If that advice does not resolve the problem, a more active intervention may be required. This involves parents imposing a graded pattern of lengthening periods between tucking their child up in bed and coming back after a few minutes to visit, but leaving the room before the child falls asleep, even if they are protesting. The object is to provide the opportunity for the child to learn how to fall sleep alone, a skill not yet

Box 23.5 Strategy for meal refusal

Mealtime history

What is the parent most concerned about?

- Nutrition?
  - Refer to growth chart
- Discipline and parenting?
  - Family history of eating problems
  - Parenting style
  - What do others say?
  - Is it part of a broader behavioural problem?

How much food is eaten between meals?

- Food diary to record child's intake over a number of days.

Advice

- As long as offered wholesome food with adequate range, children are remarkably good at eating a constant and appropriate quantity of food when allowed a reasonable choice.
- As it is impossible to force a child to eat, avoid confrontation at mealtimes.
- Develop a relaxed atmosphere.
- Use favourite foods as a reward. Introduce other rewards for compliance at mealtimes (e.g. additional privileges such as extra TV time).
- Reduce eating between meals if necessary, although many young children prefer small, frequent snacks.

developed. More refractory cases may require specialist referral.

Waking at night

This is normal, but some children cry because they cannot settle themselves back to sleep without their parent's presence. This is often associated with difficulty settling in the evenings, which should be treated first. Some children who can settle in the evening may be unable to settle when they wake in the night because the circumstances are different – it is quieter, darker, etc. The graded approach described above for evening settling can also be used in the middle of the night. Parents will find it helpful to take alternate nights on duty to share the burden.

Nightmares

These are bad dreams which can be recalled by the child. They are common, rarely requiring professional attention unless they occur frequently or are stereotyped in content, indicating a morbid preoccupation or symptomatic of a psychiatric disorder such as post-traumatic stress disorder. Unless a disorder is suspected, reassuring the child and his family will usually suffice.

Night (sleep) terrors

These are different from nightmares, occurring about 1.5 hours after settling. The parents find the child sitting up in bed, eyes open, seemingly awake but

Box 23.6 Reasons for a child not settling at night

- Too much sleep in the late afternoon
- Displaced sleep/wake cycle – not waking child in morning because did not settle until late on the previous night
- Separation anxiety
- Overstimulated or overwrought in the evening
- Kept awake by siblings or noisy neighbours or TV in the bedroom
- Erratic parental practices: no bedtime or routine to cue child into sleep readiness, sudden removal from play to go to bed without prior warning to wind down
- Use of bedroom as punishment
- Dislike of darkness and silence – night light and playing story tapes can be helpful
- Some chronic physical conditions may be associated with sleep problems, e.g. painful crisis in sickle cell disease.

obviously disorientated, confused and distressed and unresponsive to their questions and reassurances. The child settles back to sleep after a few minutes and has no recollection of the episode in the morning. A night terror is a *parasomnia*, a disturbance of the structure of sleep wherein a very rapid emergence from the first period of deep slow-wave sleep produces a state of high arousal and confusion. Sleepwalking has similar origins and the two may be combined. Most night terrors need little more than reassurance directed towards the parents. The most important intervention for sleepwalking is to make the environment safe to prevent injury to the child (e.g. not sleeping on the upper bunk of a double-bunk bed, putting gates before the staircase, locking the kitchen, etc.). Given that a common cause of night terrors and sleepwalking is a poor and erratic sleep schedule, a sleep routine can be helpful in preventing recurrence. Once parents have implemented the safety suggestions highlighted above, they can be reassured, as the natural course of these disorders is to decrease over time.

Disobedience, defiance and tantrums

Normal toddlers often go through a phase of refusing to comply with parents' demands, sometimes angrily ('the terrible 2s'). This is an understandable reaction to the discovery that the world is not organised around them. They also become confused and angered by the fact that the parent who provides them with comfort when they are distressed is also the person who is making them do things they do not wish to do. This seems exceptionally unfair to them. That is one reason why children play their parents up but may be fine with others. All this can exhaust and demoralise parents, not least because many people offer advice or criticism (everyone thinks themselves an expert in the area of children's development and behaviour). The points listed in Box 23.7 can be made.

Box 23.7 Managing toddler disobedience

- Ensure your demand is reasonable for the developmental stage of the child
- Tell the child what you want him/her to do rather than nagging about what you do not want him/her to do
- Praise for compliance, especially when it is spontaneous (catch doing the right thing)
- Use simple incentives to reward good behaviour
- Use instructions like 'If you (do this or that) … then we/I can do such and such' (not the other way round)
- Avoid threats that cannot be carried out
- Follow through with any consequences you indicated for non-compliance
- Ignore some episodes of defiance if they are not significant

Box 23.8 Analysing a tantrum

- **A**ntecedents – what happened in the minutes before the episode
- **B**ehaviour – exactly what did the episode consist of
- **C**onsequences – what happened as a result, including what you did and the outcome.

Temper tantrums are ordinary responses to frustration, especially at not being allowed to have or do something. They are common and normal in young preschool children. If asked for advice, a sensible first move is to take a history, analysing a couple of tantrums according to the ABC paradigm (Box 23.8). Next, examine the child to identify potential medical or psychological factors. Medical factors include global or language delay, hearing impairment (e.g. glue ear) and medication with bronchodilators or anticonvulsants. If none are present, there are management strategies that can be adopted, some of which are shown in Box 23.9.

The easiest course of action is to distract the child or, if this cannot be done, to let the tantrum burn itself out while the parent leaves the room, returning a few minutes later when things quieten down (provided it is safe to leave the child alone). Obviously this should be done in a calm, neutral manner and certainly not accompanied by threats of abandonment. Tantrums which are essentially coercive (when a child is demanding something from a parent) must be met by a refusal to give in. They can often be forestalled by the simple expedient of making rules which the child can be reminded of before the situation presents itself. An alternative course is to use 'time out', which is a form of structured ignoring. The child in a tantrum is placed somewhere such as the hallway, where no-one will talk to him for a short time, e.g. 1 min per year of age. During this period they are ignored completely. Parents often expect this manoeuvre to produce a contrite child, complaining if it does not do so immediately. In fact, when used for tantrums, time out works according to different principles (not as a response to punishment but to the withdrawal of attention) and often takes several weeks to effect a gradual improvement. It may help to ask the mother to keep records to document this.

Box 23.9 Tantrums: management strategies

- Affection and attention before the tantrum
- Distraction
- Avoiding antecedents
- Ignoring:
  - Effective but can be difficult
  - No surrender (when parents give in, tantrums become harder to deal with over time)
- Time out from positive reinforcement:
  - Walk away, returning when quietens down
  - Separate from siblings
  - Put on a 'naughty chair' for a short time
- Holding firmly if the child is putting themselves or others in danger
- Star chart to prevent future episodes.

Disobedience can be dealt with by using a star chart to reward the child for complying with parental requests. The chart needs to be where the child can see it and it must be the case that the child knows what to do in order to get a star. It is wisest not to 'fine' the child by taking stars away once they have been earned. If the parent who is rewarding compliance by the child praises at the same time as giving the star, there may not be the need to tie stars with a material reward. However, if a tangible reward had been promised for a certain number of stars, it is important to follow through with this.

For temper tantrums

- **Analyse according to antecedents, behaviour and consequences**
- **Consider distraction, avoiding antecedents, ignoring and time out.**

Aggressive behaviour

Small children can be aggressive for a host of reasons, ranging from spite to exuberance. Much aggressive behaviour is learned, either by being rewarded (often inadvertently) or by copying parents, siblings or peers. For example, many instances of aggressive, demanding behaviour are provoked or intensified by a parent shouting at or hitting their child. In such cases, it is the parent's behaviour which needs to change. In most instances, the same principles as apply to tantrums are valid: make rules clear, stick to them, keep cool, do not give in and use time out if necessary. The latter can often be used on a 1–2–3 principle (Fig. 23.4). A tired or stressed child will be irritable and prone to

The 1–2–3 principle for tantrums or aggressive behaviour

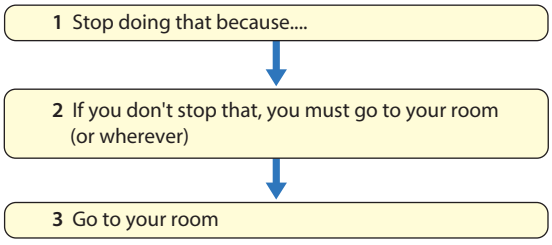


Figure 23.4 The 1–2–3 principle for tantrums or aggressive behaviour.

angry outbursts, as will children whose communication skills are compromised by deafness or a developmental language disorder so that they are frustrated and exasperated. Optimistic reassurance that the child will spontaneously grow out of a pattern of aggressive behaviour is mistaken; once established, an aggressive behavioural style is remarkably persistent over a period of years. Thus, aggressive behaviour in children needs to be proactively managed. There are several evidence-based parenting programmes that are effective for teaching parents to manage aggression in their children. Parents should be encouraged to attend such programmes.

Autism

This is considered in Chapter 4.

Problems of middle childhood

Nocturnal enuresis

Children can wet themselves by day or night, but in colloquial speech, 'enuresis' is synonymous with bedwetting. It is quite common: about 6% of 5-year-olds and 3% of 10-year-olds are not dry at night. Boys outnumber girls by nearly 2 to 1. There is a genetically determined delay in acquiring sphincter competence, with two-thirds of children with enuresis having an affected first-degree relative. There may also be interference in learning to become dry at night. Small children need reasonable freedom from stress and a measure of parental approval in order to learn nighttime continence. It is well recognised that emotional stress can interfere and cause secondary enuresis (relapse after a period of dryness). Most children with enuresis are psychologically normal and the treatment of secondary enuresis still relies mainly on the symptomatic approach described below, although any underlying stress, emotional or physical disorder must be addressed.

Organic causes of enuresis are uncommon but include:

- Urinary tract infection
- Faecal retention severe enough to reduce bladder volume and cause bladder neck dysfunction
- Polyuria from osmotic diuresis, e.g. diabetes mellitus, or renal concentrating disorders, e.g. chronic renal failure.

A urine sample should always be tested for glucose and protein and checked for infection. Daytime and secondary enuresis are considered in Chapter 18.

The management of nocturnal enuresis is straightforward but needs to be painstaking to succeed. After the age of 4 years, enuresis resolves spontaneously in only 5% of affected children each year. In practice, treatment is rarely undertaken before 6 years of age.

Explanation

The first step is to explain to both child and parent that the problem is common and beyond conscious control. The parents should stop punitive procedures, as these are counterproductive.

Star chart

The child earns praise and a star each morning if the bed is dry. Wet beds are treated in a matter-of-fact way and the child is not blamed for them.

Enuresis alarm

If a child does not respond to a star chart, it may be supplemented with an enuresis alarm. This is a sensor, usually placed in the child's pants or under the child, which sounds an alarm when it becomes wet. In order to be effective, the alarm must wake the child, who gets out of bed, goes to pass urine, returns and helps to remake a wet bed before going back to sleep. It is not necessary to reset the alarm that night. Parental help can be enlisted in the night using a baby alarm to transmit the noise of the alarm to the parents' bedroom.

The alarm method takes several weeks to achieve dryness but is effective in most cases so long as the child is motivated and the procedure is followed fully. About one-third relapse after a few months, in which case repeat treatment with the alarm usually produces lasting dryness.

Desmopressin

Short-term relief from bedwetting, e.g. for holidays or sleepovers, can be achieved by the use of the synthetic analogue of antidiuretic hormone, desmopressin, taken as tablets or sublingually. This achieves a suppressant effect rather than a lasting cure.

Self-help groups

These provide advice and assistance to parents and health professionals, e.g. the Enuresis Resource and Information Centre (ERIC).

Summary

Nocturnal enuresis

- Common, males more than females
- Most affected children are psychologically and physically normal
- Treatment usually considered only at >6 years of age
- Management – explanation, star charts, enuresis alarm, sometimes desmopressin.

Faecal soiling

It is abnormal for a child to soil after the age of 4 years. Thereafter, children who soil fall into two broad groups: those with and those without a rectum loaded with faeces. Because of this, it is important to ascertain whether there is faecal retention by abdominal palpation. The reasons why a child's rectum becomes loaded are various, and commonly involve an interplay between constitutional factors and experience. Some children have a rectum that only empties occasionally, perhaps because of poor coordination with anal sphincter relaxation, and are thus more prone to developing retention. Superimposed upon this are a number of other factors:

- Constipation, possibly following dehydration during an illness
- Inhibition of defecation because of pain from a fissure
- Inhibition because of fear of punishment for incontinence
- Anxieties about using the toilet.

Once established, a huge bolus of hard faeces may be beyond the capacity of the child to shift. Furthermore, a rectum loaded with hard or soft faeces (both are found) dilates and habituates to distension so that the child becomes unaware of the need to empty it. The loaded rectum inhibits the anus via the rectoanal reflex and stool may seep out with spontaneous rectal contractions beyond the child's control. Soiling occurs in the child's pants, which may then be removed and hidden out of shame.

Any reasons for faecal retention, such as an anal fissure, should be identified and treated, but the most important thing is to empty the rectum as soon as possible. The child and parents need to understand that retention is present and how it leads to incontinence.

A stool softener (macrogol) is given for a couple of weeks, followed, if necessary, by a stimulant laxative (docusate, sodium picosulphate or senna) and an osmotic laxative (lactulose). Sometimes an enema is required. Once the rectum is disimpacted, maintenance laxative therapy is maintained (see Fig. 13.14 for details). The child can be encouraged to defecate regularly in the toilet, which earns stars on a star chart. Such retraining may take a number of weeks while the distended rectum shrinks to a normal size. Throughout this period a regular laxative is usually needed. The stars may therefore need to be cashed in for tangible rewards such as extra pocket money in order to maintain the incentive.

In some cases, repeated soiling will have been such a humiliating experience for the child that they psychologically deny there is a problem and cooperation is doubtful. Other children find that their involuntary soiling allows them a measure of control over parents and they are reluctant to surrender an apparently useful weapon. Such cases may need psychiatric referral.

Soiling may occur in conjunction with an empty rectum for various other uncommon reasons. Some children have an urgency of defecation for apparently constitutional reasons and can only postpone defecation for a few minutes; they can be taken by surprise. Some children have neuropathic bowel secondary to occult spinal abnormality, usually associated with urinary incontinence. Similarly, diarrhoea can overwhelm bowel control. The child may have a general learning disability with a mental age below 4 years, so that expectations of social bowel control need to be revised accordingly. Lastly, the child may defecate intentionally as a hostile act. Such children may be entrenched in distorted relationships with their parents and may have other behavioural problems requiring psychiatric referral.

Summary

Faecal retention

- Present in most children who soil
- May be due to constipation or reluctance to open the bowels because of pain or reluctance to use the toilet
- When present, the rectum needs to be emptied, initially with a stool softener and laxative, followed by retraining.

Recurrent unexplained somatic symptoms/somatisation

Recurrent medically unexplained (functional somatic) symptoms are common in childhood and adolescence. In many cases, they are aggravated by stress but they can also be the expression of an anxiety or depressive disorder. Somatisation is the term used for the communication of emotional distress, troubled relationships and personal predicaments through bodily symptoms. The prepubertal child may experience affective distress as recurrent abdominal pain (this symptom peaking at age 9 years) and headaches (peaking at age 12 years). With increasing age, limb pain, aching muscles, fatigue and neurological symptoms become more prominent.

Recurrent central abdominal pain, often sharp and colicky, affects about 10% of school-age children. The causes are considered in Chapter 13. In the majority of cases, no organic cause can be objectively demonstrated, yet the child is obviously in pain. Some will have an emotional cause for their pain, but in many, no aetiology, medical or psychiatric, can be demonstrated.

The history must attend to possible sources of stress and the child should be interviewed about school, friends and family, noting the general level of anxiety and ability to communicate. This should be an integral part of the interview and not done as an afterthought when organic causes have been excluded. A thorough physical examination is important to reassure the child and family that there is no underlying organic cause. It also provides an opportunity to gain further information about the nature of the pain and the child's reaction to it. When examining the child, it is sensible to ask the child to point to where the pain is. In general, the further the pain is from the umbilicus, the more

likely it is being caused by organic pathology (Apley's rule).

The pain may be limited to school days or coincide with upsetting events in the home, such as parental conflict, or other specific situations. A short interview with the child on their own can reveal sources of stress which may be otherwise unrecognised by parents or which the child is wary of mentioning in front of them. Problems at school, particularly bullying and teasing, or difficulties with a teacher or class work may only be known by the child. A report from the school may be helpful. A joint interview with both parents and the child is a good arena for explaining to the child and family how organic disease has been ruled out and, if appropriate, how tension can give rise to pain using familiar examples such as headache. It is often necessary to promote communication between family members to avoid any tendency for somatic symptoms to replace verbal communication of distress. Learning pain-coping skills, such as relaxation, may be helpful, especially for headaches. Referral to child and adolescent mental health services is indicated if any identified stressors cannot be relieved by straightforward means, if there is serious family dysfunction, or if the pain impairs the child's general functioning at home or school.

Summary

Somatic symptoms

- May be a means of communicating emotional distress
- Sources of stress should be identified, and ameliorated if possible
- In many children with unexplained recurrent abdominal pain or headaches, no significant sources of stress are identified.

Tics

A tic is a quick, sudden, coordinated movement, which is apparently purposeful and recurs in the same part of the child's body. It is not entirely involuntary in that it can be purposefully suppressed to some extent. About 1 in 10 children develop a tic at some stage, typically around the face and head – blinking, frowning, head-flicking, sniffing, throat clearing and grunting being the commonest. They are most likely to occur when the child is inactive (watching TV or on long car journeys) and often disappear when actively concentrating. They may worsen with anxiety but they are not themselves an emotional reaction. In most cases, there is a family history. These simple, *transient childhood tics* clear up over the next few months, although they may recur from time to time. They should be treated with reassurance in the first place.

Less commonly, the child has tics from which he/she is hardly ever free. They may be multiple, although there is fluctuation in the predominance of any particular tic and in overall severity. This is a chronic tic disorder which, if it includes both multiple motor tics and vocal tics such as hooting, yelping or swearing, is known as *Gilles de la Tourette's* syndrome. These conditions tend to be persistent in the medium term, requiring medication (such as clonidine or risperidone) under specialist supervision.

Hyperactivity

Young children are characteristically lively, some more than others, by virtue of their immaturity. When their level of motor activity exceeds that regarded as normal, they may be termed 'hyperactive' by their parents. This is a judgement that depends upon the parents' standards and expectations. The term can thus incorrectly be used as a complaint about a child who is normally active in overall terms but who can be cheeky and boisterous at times. Such a child is *not* hyperactive, but the parents need advice about how to handle unwanted behaviour.

In the true *hyperkinetic disorder* or *attention deficit hyperactivity disorder* (ADHD), the child is undoubtedly overactive in most situations and has impaired concentration with a short attention span or distractibility. Differences in diagnostic criteria and threshold mean that prevalence rates among prepubertal schoolchildren are variously estimated as between 10 and 50 per 1000 children, boys exceeding girls three-fold. There is a powerful genetic predisposition and the underlying problem is a dysfunction of brain neuron circuits that rely on dopamine as a neurotransmitter and which control self-monitoring and self-regulation.

Affected children are unable to sustain attention or persist with tasks. They cannot control their impulses – they manifest disorganised, poorly-regulated and excessive activity; have difficulty with taking turns; sharing; are socially disinhibited; and butt into other people's conversations and play. Their inattention and hyperactivity are worst in familiar or uninteresting situations. They also cannot regulate their activity according to the situation – they are fidgety; have excessive movements inappropriate to task completion; lose possessions; and are generally disorganised. Typically, they have short tempers and form poor relationships with other children, who find them exasperating.

The children do poorly in school and lose self-esteem. They may drift into antisocial activities for a variety of reasons, partly because their behaviour drives parents, teachers and peers to use coercion and punishment, which is ineffectual or breeds resentment.

In addition to child psychiatric or paediatric evaluation, the child will usually need to be assessed by an educational psychologist.

First-line management in preschool children and school-aged children with mild to moderately severe disorder is the active promotion of behavioural and educational progress by specific advice to parents and teachers to build concentration skills, encourage quiet self-occupation, increase self-esteem and moderate extreme behaviour. Behavioural interventions similar to those embedded in parenting programmes are helpful. These involve having clear rules and expectations, and consistent use of rewards to encourage adherence and where appropriate, consequences to discourage unacceptable behaviour.

For those children in whom this is insufficient, hyperactivity responds symptomatically to several types of medication, although this is usually reserved for children older than 6 years of age. Stimulants such as methylphenidate or dexamphetamine and non-stimulants, like atomoxetine, reduce excessive motor activity and improve attention on task, focused behaviour. The usual approach is not to put the child on medication until behavioural and educational progress is actively promoted by the specific measures mentioned above. However, in severe cases with high degrees of impairment, simultaneous psychosocial and medical treatment may be required. It may be necessary to continue medication for several years, sometimes into adulthood. Yearly trial off medication is recommended to evaluate the need for continuing treatment. Specialist supervision is mandatory. Close liaison with the school is required throughout the years of treatment.

The role of diet in the cause and management of hyperactivity is controversial. Current evidence indicates that the sort of diet which aims blindly to reduce sugar, artificial additives or colourants has no effect. A few children display an idiosyncratic behavioural reaction such as excitability or irritability to particular foods. If this seems likely, trying the child on an exclusion of that particular food may be useful. In general, food and drinks with caffeine are not advised. Overzealous dietary exclusion can lead to malnutrition, especially in a child on stimulant medication that may already have the side-effect of appetite reduction.

Summary

Attention deficit hyperactivity disorder (ADHD)

- Affects males more than females
- Clinical features: cannot sustain attention, excessively active, socially disinhibited, easily distracted and impulsive, may be poor at relationships, prone to temper tantrums, poor school performance
- Management: educational psychologist assessment, behavioural programmes in school, parenting intervention, medication if necessary.

Antisocial behaviour

Children steal, lie, disobey, light fires, destroy things and pick fights for various reasons:

- Failure to learn when to exercise social restraint
- Lack of social skills, such as the ability to negotiate a disagreement
- They may be responding to the challenges of their peers in spite of their parents' prohibitions
- They may be chronically angry and resentful
- They may find their own notions of good behaviour overwhelmed by emotion such as sadness or temptation.

When serious antisocial behaviour which infringes the rights of others is the dominant feature of the clinical picture and is so severe as to represent a handicap to general functioning, a diagnosis of *conduct disorder* is made. Children with conduct disorder may not have necessarily broken the law, although their behaviour excites strong social disapproval. They typically come from homes in which there are considerable discord, coercive relationships, limited boundaries that are inconsistently enforced, and poor supervision by adults. A milder form, characterised by angry, defiant behaviour to authority figures such as parents and teachers, is known as *oppositional-defiant disorder* (*ODD*).

Treating conduct disorder can be difficult. Parent management training programmes (such as Webster–Stratton and Triple P) have an excellent evidence base and are highly recommended as primary interventions. However, poor parental cooperation and motivation can result in minimal benefit. Where parents are unwilling or unable to take up parenting programmes, affected children can be offered individual or group-based interventions focusing on problem-solving skills and anger management. Although these interventions show benefit in research settings, affected children do not often have the level of motivation required to benefit in routine clinic settings. In the absence of a coexisting psychiatric condition responsive to medication, it is not considered standard clinical practice to use medication for conduct disorder in the UK.

Summary

Antisocial behaviour

- It is important to exclude any coexisting psychiatric condition and treat this directly, e.g. ADHD or depression
- Parenting groups are an evidence-based treatment for these disorders, but require motivation.

Anxiety

Pathological anxiety exists in two forms: specific and general. In phobias there is fear of a specific object or situation that is excessive and handicapping and cannot be dealt with by reassurance. Most children have a number of irrational fears (the dark, ghosts, kidnappers, dogs, spiders, bats, snakes) which are common and do not usually handicap the child's ordinary life. Some of these persist into adulthood. If they are so severe that the child's ability to lead an ordinary life is affected, then treatment by cognitive behavioural therapy with graded exposure to the feared event may be indicated and is usually successful.

More diffuse general anxiety presents indirectly in childhood and it is uncommon for a child to complain directly about anxiety. Often, it is first manifest as physical complaints: nausea, headache or pain. It may take the form of health worries and the child repeatedly asks for reassurance that he is not going to die. Some children with generalised anxiety may develop unusual coping strategies that appear manipulative, in an attempt to gain control over their parents and the

world in general. It may be a justifiable reaction to an event or situation, or be disproportionate. If the condition follows a recognisable precipitant such as a parental illness and the parents can be directed to provide comfort and support, prognosis is good. If it arises insidiously, specialist mental health referral is indicated.

Children rarely say spontaneously that they are anxious – instead they tend to complain of aches and pains or behave in apparently manipulative ways to cope with or avoid the feared situation.

School refusal

During the years of compulsory school attendance, a child may be absent from school because of illness, because parents keep the child off school or because of truancy in which the child chooses to do something else rather than attend school. In truancy, a child leaves to go to school but never arrives or leaves early. It is often accompanied by other behavioural difficulties. A few non-attendees at school suffer from *school refusal*, an inability to attend school on account of overwhelming anxiety. Such children may not complain of anxiety but of its physical concomitants or the consequences of hyperventilation. Anxiety may present as complaints of nausea, headache or otherwise not being well, which are confined to weekday, term-time mornings, clearing up by midday. It may be rational, as when the child is being bullied or there is educational underachievement. If it is disproportionate to stresses at school, it is termed school refusal, an anxiety problem with two common causes – separation anxiety from parents persisting beyond the toddler years and anxiety provoked by some aspect of school, true school phobia. These can coexist.

School refusal based on separation anxiety is typical of children under the age of about 11 years. It may be provoked by an adverse life event such as illness, a death in the family or a move of house. The child is unable to tolerate separation from their attachment figure without whom the child cannot go anywhere, including school. Treatment is aimed at gently promoting increasing separations from the parents (e.g. staying overnight with relatives or friends), while arranging an early return to school. Some adolescents with school refusal have a depressive disorder, but more usually there is an interaction between an anxiety disorder and long-standing personality issues such as intolerance of uncertainty.

True school phobia is seen in slightly older, anxious children who are frequently uncommunicative and stubborn.

The management of school refusal is shown in Box 23.10.

Box 23.10 Treatment of school refusal

- Advise and support parents and school about the condition
- Treat any underlying emotional disorder
- Plan and facilitate an early and graded return to school at a pace tolerable for the child with all involved (child, family, teachers, educational psychologist and educational welfare officers)
- Help the parents make it more rewarding for the child to return to school than stay at home
- Address bullying or educational difficulties if present.

Box 23.11 Causes of underachievement at school

Long-standing problem

- Visual problems
- Hearing problems
- Dyslexia
- Generalised or specific learning problems
- Hyperactivity
- Anti-education family background
- Chaotic family background

Recent onset of problem

- Preoccupations (parental divorce, bullying, etc.)
- Fatigue
- Depression
- Rebellion against teacher, parents or 'swot' label
- Unsuspected poor attendance at school
- Sexual abuse
- Drug abuse
- Prodromal period of a psychotic illness (rare)
- Degenerative brain condition, rare but important.

Educational underachievement

Children who achieve less well in school than expected are sometimes brought to doctors. It is important to evaluate parents' and teachers' expectations and ensure the child is actually able to rise to them. The services of an educational psychologist are indispensable. Core medical responsibilities include testing sight and hearing and attempting to elicit the cause of underachievement according to the list in Box 23.11. The topic is considered further in Chapter 4.

Adolescence

Although a popular image of adolescence is one of angry, rebellious teenagers, alienated from their parents and embroiled in emotional turmoil, studies show that most adolescents maintain good relationships with their parents. They do, however, tend to bicker with them about minor domestic matters and what they are allowed to do. Minor psychological symptoms such as moodiness or social sensitivity are quite common (as they are in adults), but serious psychiatric problems are no more prevalent than in adult

Box 23.12 Formal operational thought

- The ability to form abstract thoughts
- Comparing implications of hypotheses
- Thinking about one's own thinking
- Testing the logic that links propositions
- Manipulating interactive abstract concepts.

life. Family relationships are often influenced by teenagers' negotiation of their own autonomy, the emergence of their own sense of themselves and the first moves towards a personal identity. At the same time, their parents may be experiencing mid-life crises of confidence in career, physical appearance or sexuality, so that parental and teenage preoccupations coincide, not always helpfully.

Cognitive style

The style of thought specifically associated with adolescence is formal operational (abstract) thought (Box 23.12), but this is acquired at various ages by different individuals during the teenage years, and a substantial minority do not develop it at all. Doctors are at a disadvantage here, as they have been selected by a series of examinations for excellence of their ability to manipulate abstractions and compare hypothetical predictions; they have often forgotten what it is like to think otherwise and communicate poorly with patients who still think concretely and practically (school-age children, about half of all teenagers and perhaps 1 in 5 adults). When interviewing adolescents, the skill is to avoid being patronising, while being sensitive as to whether abstract and reflective thought is solidly achieved. Using practical examples (not metaphors) and checking whether you have been understood will help to avoid the common problem of being faced with an adolescent who responds to questions with a sullen 'don't know'. This is considered further in Chapter 28.

Anorexia nervosa

Dieting to slim is endemic among teenage girls. Part of the reason for this is the contemporary equation between thinness and attractiveness, an assumption prevalent in advertising and fashion. Resonant with this is the finding that most teenage girls (but very few boys) overestimate their body width and depth, perceiving and judging themselves as fatter than they actually are.

Slimming through self-imposed calorie restriction is usually self-limiting because the goal is achieved or because the girl gives up; hunger wins through. In some girls, however, the slimming process takes over and there supervenes what has been called a 'relentless pursuit of thinness', typically with a phobic horror of normal body weight and shape. This is *anorexia nervosa*, and the features are:

- Self-induced weight loss resulting in a low body mass index (BMI); in children this needs to be plotted on a BMI centile chart, in older adolescents it is ≤17.5 kg/m^2
- A distorted perception of her body, which increases with weight loss
- A determined attempt to lose weight or avoid weight gain, by either restricting food intake, self-induced vomiting, laxative abuse, excessive exercising or using a combination of these methods
- When body weight falls below a critical point, pubertal development is halted and reversed so that menstruation ceases and the girl effectively becomes a prepubertal child. This may spare her some of the challenges of adolescence, particularly those related to sexuality
- The discovery by a girl who has felt powerless that through self-starvation she can control her shape and development and thus increase her sense of self-worth and self-effectiveness
- Preoccupations and dreams of food and cooking which come to dominate mental life as a response to starvation. There ensues a tremendous mental struggle not to give in and eat, which assumes prime importance in the girl's mental life
- The dramatic and visible effects of self-starvation on the girl, which can unite some parents in caring for their daughter and save a discordant marriage from divorce, something which she may fear is imminent.

An affected girl will often deny hunger, reassure everyone that she is in the peak of health, exercise to lose weight and disagree fervently that she is too thin. She will be careless of her own emaciation and seem unconcerned that she is starving herself to death. To the bewilderment of her parents, she may cook for others and read cookery books avidly. She may well be deceitful to anyone she perceives as thwarting her in her quest. Thus, she will conceal her poor eating by secretly disposing of her meals or lying about her weight. Both before and during her illness she will show obsessional, perfectionist character traits; without these she would not have the capacity to establish herself as a persistent dieter. Indeed, she is likely to be described as having been quiet, compliant and hard-working, 'the last person to develop anorexia nervosa'. Her parents will often present as nice people who avoid conflict.

As a result of starvation, her body develops a low metabolic rate with slow-to-relax tendon reflexes, reduced peripheral circulation, bradycardia and amenorrhea. Fine lanugo hair appears over her trunk and limbs. She does not lose pubic or axillary hair, although incompletely established puberty is delayed. Serum T_3 (triiodothyronine) may be low, giving rise to a false suspicion of hypothyroidism. Plasma proteins are sometimes low and ankle oedema not uncommon. Blood and urine levels of luteinising hormone and follicle-stimulating hormone are low and non-cyclical.

Some girls discover that self-restraint in carbohydrate intake can be bypassed by self-induced vomiting following repeated bouts of overeating and that further weight loss can be achieved by diuretics, and laxatives (in the belief that these will expedite food transit time and reduce absorption). This can cause

wide fluctuations in weight and metabolic abnormalities such as hypokalaemia and alkalosis. This condition is *bulimia* which can occur at normal body weight or in association with low body weight as an ominous complication of anorexia nervosa. It tends to affect older rather than younger teenagers. Bulimia at normal body weight can be managed by encouraging a regular diet, monitoring this by a diary and providing individual or group cognitive behavioural therapy.

The prevalence rate among teenagers for anorexia nervosa is a little less than 1%, but the incidence rate may have increased over the last 50 years. The peak age of onset is 14 and girls outnumber boys by about 10:1. Bulimia is commoner, although prevalence rates vary widely, depending on the degree of severity. It also shows a markedly female preponderance and may also be becoming more frequent.

Management

Management is two-fold: medical and psychological. The initial management of anorexia nervosa is to restore near-normal body weight by refeeding. The emergence of physical complications may necessitate admission to hospital for refeeding, which may even involve nasogastric tube feeding in some instances. The cornerstone of treatment is family therapy. Individual psychological treatment is introduced to help the young person challenge the cognitions that drive anorexia and to acquire more constructive ways of confronting developmental demands, including handling conflict, maintaining self-esteem, personal autonomy and relationships.

Medical aspects

Anorexia has a high mortality rate compared with other psychiatric disorders. Some of the excess mortality arise from medical complications such as malnutrition, electrolyte imbalance and infection. This emphasises the importance of thorough physical examination, investigations and medical management. In the UK, NICE (National Institute for Health and Clinical Excellence) has produced a guideline for treatment of eating disorders including the physical management of anorexia nervosa.

Prognosis

The prognosis for children and adolescents is variable, with as many as 50% failing to make a full recovery. Factors predicting a poorer outcome include a low BMI and physical complications prior to treatment, bulimic symptoms, especially self-induced vomiting, as well as family disturbance, and interpersonal difficulties. Anorexia has the highest mortality of all psychiatric disorders. In addition to medical complications, the next important cause of mortality is suicide.

Summary

In anorexia nervosa

- Female:male ratio is 10:1
- Peak age of onset – 14 years
- Affected girls have a distorted body image, so seldom agree that they are too thin and may deceive everyone by pretending to eat
- Features include: determined efforts to lose weight, arrest of puberty, cessation of periods
- May be accompanied by bulimia: overeating followed by self-induced vomiting
- Management is family therapy and individual therapy to restore body weight
- Some require hospitalisation; prognosis is variable, but has a mortality from suicide, malnutrition and infection.

Chronic fatigue syndrome

Chronic fatigue syndrome (CFS) refers to persisting high levels of subjective fatigue, leading to rapid exhaustion on minimal physical or mental exertion. The term is broader and more neutral than the specific pathology or aetiology implied by myalgic encephalomyelitis (ME) or post-viral fatigue syndrome, which follows an apparently viral febrile illness. There is sometimes serological evidence of recent infection with coxsackie B or Epstein–Barr virus (EBV) or a hepatitis virus. Some cases have no history or evidence of a precipitating infection and there are no specific diagnostic tests. The clinical picture is somewhat diffuse and there are no pathognomonic symptoms. Myalgia, migratory arthralgia, headache, difficulty getting off to sleep, poor concentration and irritability are virtually universal. Stomach pains, scalp tenderness, eye pain and photophobia, and tender cervical lymphadenopathy are frequently encountered. Depressive symptoms are common and there is continuing debate as to how much of the clinical picture is physical and how much psychological. Usually parents insist on there being a physical cause and there is a risk that the doctor will carry out excessive unnecessary investigations. Most experienced doctors now regard the final clinical picture as resulting from both physical and psychological factors.

The majority of cases will remit spontaneously with time, but this takes months or sometimes years. Earlier recommendations of continuous rest have been shown to be unhelpful and can lead to secondary complications. The recommended treatment involves graded exercise therapy and/or cognitive behavioural therapy. Graded exercise therapy is usually provided by physiotherapists and aims to achieve gradual increase in exercise tolerance. If too much pressure is put upon the

Summary

In chronic fatigue syndrome

- There is exhaustion on minimal exertion
- There is thought to be a combination of physical and psychological factors
- Management is with graded exercise and/or cognitive behavioural therapy, but recovery may take months or years.

child, tantrums or mute withdrawal can occur. Argument about how much of the condition is physical and how much psychological is unhelpful. The parents and the child need continuing support to maintain as much of a normal life as possible, including school attendance. The mood of children with depressive symptoms may respond to antidepressant medication, but this is a treatment only for depressive symptoms and it is unlikely to result in alleviation of the fatigability. NICE guidelines are available.

Depression

Low mood can arise secondary to adverse circumstances or sometimes spontaneously. Depression as a clinical condition is more than sadness and misery; it extends to affect motivation, judgement, the ability to experience pleasure and provokes emotions of guilt and despair. It may disturb sleep, appetite and weight. It leads to social withdrawal, an important sign. Such a state is well recognised among adolescents, particularly girls, but occasionally affects prepubertal children. The general picture is comparable to depression in adults but there are differences (Box 23.13).

A diagnosis of depression depends crucially upon interviewing the adolescent on his own, as well as taking a history from the parents. Teenagers will, out of loyalty, often pretend to their parents that things are all right if interviewed in their presence. It is necessary to ask about feelings directly and to ask specifically about suicidal ideas and plans.

Treatment depends upon severity. Children with mild depression are managed initially in primary care and other non-specialist mental health settings. Many will recover spontaneously; hence a period of watchful waiting for up to 4 weeks may be appropriate. Alternatively, the child could be offered non-directive supportive therapy or guided self-help. However, if mild depression does not respond to these measures in 2–3 months, the child should be referred to specialist mental health services. Similarly, children with moderate and severe depression should be referred to specialist mental health services for more specific psychological intervention such as cognitive behavioural therapy, family therapy or interpersonal therapy. In all cases, any identified contributing factor such as bullying needs to be addressed. If psychological therapy for moderate or severe depression is insufficient after 6 weeks, then an SSRI (selective serotonin reuptake inhibitor antidepressant), fluoxetine, should be considered. Depressed young people who are suicidal may need admission to an adolescent psychiatric in-patient unit.

Box 23.13 Features of depression in adolescents

More common than adults

- Apathy, boredom and an inability to enjoy oneself rather than depressed mood
- Separation anxiety which reappears, having resolved in earlier life
- Decline in school performance
- Social withdrawal
- Hypochondriacal ideas and complaints of pain in chest, abdomen and head
- Irritable mood or frankly antisocial behaviour

Less common than adults

- Loss of appetite and weight
- Loss of sleep
- Loss of libido
- Slowing of thought and movement
- Delusional ideas.

Self-harm

Like adults, young people who take overdoses do so for a variety of motives, of which suicide is the most serious one. For a high proportion, the overdose is a desperate gesture which may draw attention to a predicament perceived by them as irresolvable. Issues such as bullying or abuse should be considered. About half of teenagers who overdose are clinically depressed. Episodes of self-harm must be taken seriously as they carry significant risk of recurrence and suicide.

In those who have taken an overdose, a full psychiatric assessment is needed by a mental health professional. A useful adjunct in the assessment of suicide risk is the PATHOS score shown in Box 23.14; however, this needs to occur alongside a psychiatric assessment.

Drug misuse

Most teenagers are exposed to illicit drugs at some stage. A number will then experiment with them, some becoming habitual users. Usually, this is for recreational purposes, but a few use them to avoid unpleasant feelings or memories. A very small number become dependent, psychologically or physically. What is taken varies with culture and opportunity but alcohol and cannabis are common; solvents, LSD, ecstasy and amphetamine derivatives somewhat less so; and cocaine or heroin currently least prevalent, though their use is increasing. The addictive potential of the

Box 23.14 PATHOS instrument to assess suicide risk after adolescent overdose

- **P:** Have you had **P**roblems for longer than a month?
- **A:** Were you **A**lone in the house at the time?
- **T:** Did you plan the overdose for longer than **T**hree hours?
- **HO:** Are you feeling **Ho**peless about the future?
- **S:** Were you feeling **S**ad for most of the time before the overdose?

Score 1 for Yes; 0 for No and add together. Child at high risk if score >2. However, the final judgement of suicide risk is a clinical and qualitative decision, not one based on a cut-off score.

Source: Kingsbury S. 1996. PATHOS: a screening instrument for adolescent overdose: a research note. *Journal of Child Psychology and Psychiatry* 37:609–611.

last two is the greatest and their dangers are well known.

Abuse implies heavy misuse. The signs vary with the agent but may include:

- Intoxication
- Unexplained absences from home or school
- Mixing with known users
- High rates of spending or stealing money
- Possession of the equipment required for drug use
- Medical complications associated with use.

Doctors may be approached by parents worried that their adolescent child may be abusing drugs. An assessment will involve interviewing the adolescent, possibly combined with taking a urine sample for drug screening. Most areas have specific services for adolescents with drug and/or alcohol problems. These services usually take self-referrals so that young people with these difficulties can access them directly. Medical involvement is predominantly focused on users who have other psychopathology including depression, or with the physical consequences of intoxication or injection when these threaten health. Solvent abuse (mainly glue and aerosol sniffing) is quite widespread as a group activity of young adolescents in some areas. It can occasionally give rise to cardiac dysrhythmias, bone marrow suppression or renal failure, and any of these can cause death, as may a fall or road traffic accident when intoxicated. Cannabis and LSD use may trigger anxiety or psychotic disorders. Ecstasy taken at dances or raves can cause dangerous hyperthermia, dehydration and death.

Doctors need to ensure that any adolescent known to them who is thought to be using drugs knows the specific risks to health. Dependence is rare among teenagers and most likely to involve alcohol. The few who are using illicit drugs for respite from psychological distress need referral to a psychiatrist.

Psychosis

Psychosis is a breakdown in the perception and understanding of reality and a lack of awareness that the person is unwell. This can affect ideas and beliefs, resulting in delusional thinking where abnormal beliefs are held with an unshakeable quality and lead to odd behaviour. The connectedness and coherence of thoughts may break down, so that speech is hard to follow, leading to thought disorder. Perceptual abnormalities lead to hallucinations, where a perception is experienced in the absence of a stimulus.

Psychotic disorders include:

- Schizophrenia, where no specific medical cause is identified and there is generally no major disturbance of mood other than blunting or flattening of affect
- Bipolar affective disorder, where the psychosis is associated with lowered mood as in depression or elevation in mood as in mania
- Organic psychosis occurs in delirium, substance-induced disorders and dementia.

Both schizophrenia and bipolar affective disorder are rare before puberty, but increase in frequency of presentation during adolescence. In these disorders the psychotic symptoms occur in clear consciousness.

Investigations should include a urine drug screen, exclusion of medication-induced psychosis (e.g. high-dose stimulants or anticholinergic drugs), exclusion of medical causes (i.e. infection, seizures, thyroid abnormalities and sleep disorders) and dementia.

Where schizophrenia and bipolar disorder is suspected, urgent referral to a psychiatrist is needed for comprehensive assessment and treatment with antipsychotic medication, psycho-education, family therapy and, where appropriate, individual therapy. In the case of an organic psychosis the underlying cause needs to be treated promptly by the paediatric team, with help from mental health professionals as appropriate.

Psychosis

- **May present during adolescence**
- **May be precipitated by or be a consequence of substance abuse.**

Management of emotional and behavioural problems

For most emotional and behavioural problems, there is an interplay between adversities in the family, peer group and school and strengths or vulnerabilities in the child. Sometimes, these are referred to as risk (predisposing) factors (things that do not in themselves produce a disorder but will do so when interacting with other adversities). Conversely, they are less likely to do so if there is a compensating strength (such as high intelligence, good self-esteem, secure attachment, good peer relations or an emotionally warm relationship with a parent). An environmental adversity may be acute (a life event) or chronic. It challenges the coping skills of the child, and emotional or behavioural problems result if these are overwhelmed. The problem may resolve spontaneously or persist.

With this in mind, it is possible to talk about the three **P**s of causation:

- **P**redisposition (vulnerability)
- **P**recipitation (usually an adverse life event)
- **P**erpetuation (usually chronic stresses).

In clinical practice, a precipitant is what many people call the 'cause', but it is often the factors that perpetuate or maintain the problem that one has to deal with.

Cultural considerations

Many developed countries are increasingly ethnically diverse in relation to language, religion and culture. This diversity has many important clinical implications for child mental health. The first implication relates to the need to recognise the subgroup of young people who are refugees or asylum seekers. These children and their families have often experienced major traumatic events before arriving in their host country. They

remain highly vulnerable to mental and social-economic adversities due to past and ongoing stressful experiences.

The second implication relates to well-recognised ethnic differences in the epidemiology of some psychiatric disorders. For example, among people of African and Caribbean origin living in Western European countries, there is a clear increase in the incidence of schizophrenia but a lower incidence of anorexia nervosa compared with the indigenous Caucasian population.

Another implication of culture relates to the presentation of psychiatric symptoms. It is well recognised that the content of obsessions in children with obsessive compulsive disorder is sometimes shaped by the child's cultural and religious beliefs. This is also true of some delusions in young people with psychotic disorders. In these examples, understanding the child's religious and cultural background is essential for making an accurate diagnosis.

There are also important cultural differences about normative behaviour in children and thresholds for help-seeking. Differences in the level of stigma attached to mental illness across cultures also influence parent's help-seeking behaviour and access to child and adolescent mental health services.

The above implications suggest that cultural awareness and sensitivity are essential skills for all clinicians working or intending to work in ethnically diverse societies. The key message is that the presentation of psychiatric distress in children and young people may be coloured by their culture and language. It is therefore essential to avoid making assumptions about the significance of clinical information with cultural or religious meaning but instead to contextualise the information to the patient's culture, for example, through the use of trained interpreters.

Assessment

It is best to interview both parents if possible. While doing so, consider the quality of their relationship and the parents' mental state. Ask open questions where possible and feel able to ask directly about feelings. Assess the attitudes of the parents to the child. Obtain examples of the problem and estimate its frequency, severity, duration and the impact it has on both the child and family.

Interview the child and ask to see the older child alone as part of the assessment. Explain to the parents that you always like to have a few words with children on their own as they may have things they may feel too embarrassed to discuss with parents present. Assess the extent of the child's suffering (they may be somewhat brazen and minimise this). Keep your questions very simple and specific, making sure the child understands what it is you want to know. This also applies to teenagers. Ask about use of drugs and alcohol, experience of abuse, thoughts of self-harm and suicide. Consider whether reports from school or other involved agencies might help. In many instances, it is worth asking the parents to keep a prospective record of the problem by means of a diary or chart which you can inspect in a few days' time. Tell them what headings you want this under (such as 'antecedents, behaviour and consequences' for temper tantrums).

Treatment

Figure 23.5 shows an approach to managing a child displaying an emotional or behavioural problem. The process of making a referral to a child and adolescent mental health service (CAMHS) is most likely to succeed if the referrer has already taken some of the history and engaged the parents and child in an attempt to alleviate the problem. Many doctors, general practitioners and paediatricians in particular, are good generalists in child mental health issues and the mental health specialist should be seen as a specialist extension of their expertise, rather than a completely different sort of person.

In general, the management of children's emotional and behavioural problems:

- should be psychological rather than pharmacological
- does not need the child to be admitted to hospital (unless required as a place of safety for suicidal children or for child protection)
- involves parents as key participants
- may involve a variety of health and social service professionals.

Often more than one intervention is required and treatments are combined and several professionals become

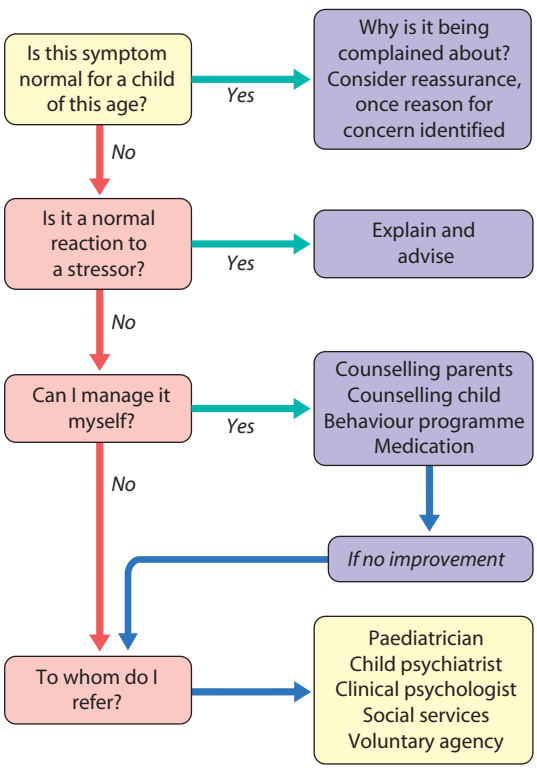


Figure 23.5 An approach to children's psychological problems.

Box 23.15 Main psychological treatment interventions employed for emotional and behavioural problems

Explanation and reassurance

Suitable for mild problems with a good prognosis arising in children from supportive families who can work out for themselves a sensible way of managing the problem until it subsides.

Counselling of child or parents

Used to provide non-directive, unstructured supportive therapy for children and families to aid coping with difficulties that are not severe enough to require specialist psychological interventions (e.g. bereavement counselling).

In parental counselling, the aim is to enhance parental coping not by telling the parent what to do but by helping them to find their own solutions, so increasing their confidence and effectiveness.

Parenting groups

Recently, parenting groups have become popular where a number of parents are seen together and given tools on how to play with their children and respond effectively to their challenging behaviour. Various approaches are rehearsed using role play and the facilitation of a therapist.

Behavioural therapy

A pragmatic approach to problems, which alters the environmental factors that trigger or maintain behaviours. It is particularly effective in the management of behavioural problems in young children.

Family therapy

Widely used by child mental health professionals. It uses a series of interviews with the entire household to alter dysfunctional patterns of relationships between family members on the basis that many children's problems are perpetuated by the ways in which family members live with and deal with each other.

Cognitive therapy

Used by specialists to explore the way thinking affects feelings and behaviour. It helps the young person to identify and challenge unhelpful thinking styles that perpetuate negative feelings and behaviour. Good evidence for efficacy in a range of disorders including depression.

Individual or group dynamic psychotherapy

More structured and intense extension of counselling, which can help children who, for example, have unconscious conflicts, which are manifest as relationship difficulties with a parent. Once the mainstay of child psychiatry, it is now less commonly used.

involved. The main treatment interventions employed are described in Box 23.15.

Medication plays a comparatively small role, although particular instances for which there is evidence for their efficacy are the use of stimulant and non-stimulant drugs in hyperkinetic disorder (ADHD), neuroleptics in psychosis and antidepressants for severely depressed adolescents. There is sometimes a temptation to sedate a child who is causing a problem but this is rarely effective and ethically questionable.

Further reading

Coghill D, Bonnar S, Duke S, et al: *Child and Adolescent Psychiatry*, Oxford Specialist Handbooks in Psychiatry, Oxford, 2009, Oxford University Press.

Goodman R, Scott S: *Child Psychiatry*, ed 2, Chichester, 2005, Wiley.

Lask B, Taylor S, Nunn K: *Practical Child Psychiatry*, London, 2003, BMJ Books.

Rutter M, Bishop D, Pine D, et al: *Rutter's Child and Adolescent Psychiatry*, Oxford, 2010, Blackwell.

24

Skin disorders

The newborn	423	Other childhood skin disorders	430
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Features of skin disorders in children include:

- In the newborn, transient skin disorders are common but need to be distinguished from serious or permanent conditions
- Atopic eczema affects up to 20% of children
- Skin infections and infestations, especially viral warts and head lice, are common in school-aged children
- Acne is troublesome for many during adolescence.

The newborn

The skin at birth is covered with vernix caseosa. This chalky-white greasy coat, mainly composed of water, proteins and lipids, protects the skin in utero from the amniotic fluid. Shedding of vernix towards the end of gestation coincides with maturation of the transepidermal barrier. In the preterm infant, the skin is thin, poorly keratinised and lacks subcutaneous fat. Transepidermal water loss is markedly increased when compared with a term infant. The preterm infant is also unable to sweat until a few weeks old, whereas the term infant can sweat from birth.

Common naevi and rashes in the newborn period are described under the examination of the newborn infant (see Ch. 9). Some less common skin conditions presenting in the newborn period are described in this chapter.

Bullous impetigo

This is an uncommon but potentially serious blistering form of impetigo, the most superficial form of bacterial infection, seen particularly in the newborn (Fig. 24.1). It is most often caused by *Staphylococcus aureus*. Treatment is with systemic antibiotics, e.g. penicillinase-resistant penicillin (see also Ch. 14).

Melanocytic naevi (moles)

Congenital moles occur in up to 3% of neonates and any that are present are usually small. Congenital pigmented naevi involving extensive areas of skin (i.e. naevi >9 cm in diameter) are rare but disfiguring (Fig. 24.2) and carry a 4–6% lifetime risk of subsequent malignant melanoma. They require prompt referral to a paediatric dermatologist and plastic surgeon to assess the feasibility of removal.

Melanocytic naevi become increasingly common as children get older and the presence of large numbers



Figure 24.1 Bullous impetigo in a 2-week-old baby.



Figure 24.2 A large (giant) congenital pigmented hairy naevus. Other smaller naevi are also visible.



Figure 24.3 A child with oculocutaneous albinism, with her parents. The hair is silvery-white.

in an adult may be indicative of childhood sun exposure. Prolonged exposure to sunlight should be avoided and sunscreen preparations with a sun protection factor exceeding 20 should be applied liberally to exposed skin in bright weather and reapplied every few hours.

Malignant melanoma is rare before puberty, except in giant naevi. However, in adults, the incidence of malignant melanoma has increased dramatically over the past 30 years. Risk factors for melanoma include a positive family history, having a large number of melanocytic naevi, fair skin, repeated episodes of sunburn and living in a hot climate with chronic skin exposure to the sun.

Parents should prevent their children becoming sunburnt.

Albinism

This is due to a defect in biosynthesis and distribution of melanin. The albinism may be oculocutaneous, ocular or partial, depending on the distribution of depigmentation in the skin and eye (Fig. 24.3). The lack of pigment in the iris, retina, eyelids and eyebrows results in failure to develop a fixation reflex. There is pendular nystagmus and photophobia, which causes constant frowning. Correction of refractive errors and tinted lenses may be helpful. In a few children, the fitting of tinted contact lenses from early infancy allows the development of normal fixation. The disorder is an important cause of severe visual impairment. The pale skin is prone to sunburn and skin cancer. In sunlight, a hat should be worn and high factor barrier cream applied to the skin.

Epidermolysis bullosa

This is a rare group of genetic conditions with many types, characterised by blistering of the skin and mucous membranes. Autosomal dominant variants tend to be milder; autosomal recessive variants may be



Figure 24.4 Severe, autosomal recessive form of epidermolysis bullosa. There is scarring following recurrent blistering.



Figure 24.5 Collodion baby.

severe and even fatal. Blisters occur spontaneously or follow minor trauma (Fig. 24.4). They need to be differentiated from scalds. Management is directed to avoiding injury from even minor skin trauma and treating secondary infection. In the severe forms, the fingers and toes may become fused, and contractures of the limbs develop from repeated blistering and healing. Mucous membrane involvement may result in oral ulceration and stenosis from oesophageal erosions. Management, including maintenance of adequate nutrition, should be by a multidisciplinary team including a paediatric dermatologist, paediatrician, plastic surgeon and dietician.

Collodion baby

This is a rare manifestation of the inherited ichthyoses, a group of conditions in which the skin is dry and scaly. Infants are born with a taut parchment-like or collodion-like membrane (Fig. 24.5). Emollients are

Box 24.1 Causes of napkin rashes

Common	Rare
• Irritant (contact) dermatitis • Infantile seborrhoeic dermatitis • *Candida* infection • Atopic eczema	• Acrodermatitis enteropathica (see p. 237) • Langerhans cell histiocytosis (see Fig. 21.18) • Wiskott–Aldrich syndrome (see p. 270).

usually applied to moisturise and soften the skin. The membrane becomes fissured and separates within a few weeks, usually leaving either ichthyotic or less commonly, normal skin.

Rashes of infancy

Napkin rashes

Napkin rashes are common, although irritant reactions are much less of a problem with the widespread use of disposable nappies, as they are more absorbent. Some causes are listed in Box 24.1. Irritant dermatitis, the most common napkin rash, may occur if nappies are not changed frequently enough or if the infant has diarrhoea. However, irritant dermatitis can occur even when the napkin area is cleaned regularly. The rash is due to the irritant effect of urine on the skin of susceptible infants. Urea-splitting organisms in faeces increase the alkalinity and likelihood of a rash.

The irritant eruption affects the convex surfaces of the buttocks, perineal region, lower abdomen and top of thighs. Characteristically, the flexures are spared, which differentiates it from other causes of napkin rash. The rash is erythematous and may have a scalded appearance. More severe forms are associated with erosions and ulcer formation. Mild cases respond to the use of a protective emollient, whereas more severe cases may require mild topical corticosteroids. While leaving the child without a napkin will accelerate resolution, it is rarely practical at home.

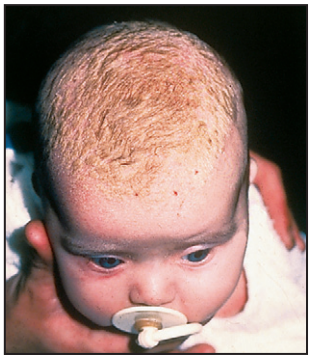
Candida infection may cause and often complicates napkin rashes. The rash is erythematous, includes the skin flexures and there may be satellite lesions (Fig. 24.6). Treatment is with a topical antifungal agent.

Infantile seborrhoeic dermatitis

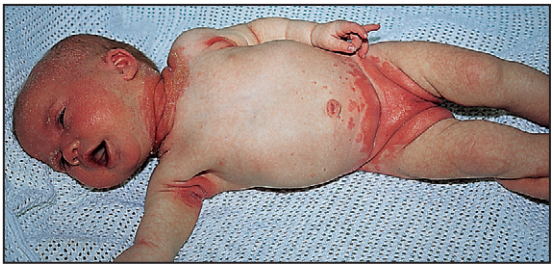
This eruption of unknown cause presents in the first 2 months of life. It starts on the scalp as an erythematous scaly eruption. The scales form a thick yellow adherent layer, commonly called cradle cap (Fig. 24.7a). The scaly rash may spread to the face, behind the ears and then extend to the flexures and napkin area (Fig. 24.7b). In contrast to atopic eczema, it is not itchy and the child is unperturbed by it. However, it is associated with an increased risk of subsequently developing atopic eczema. Mild cases will resolve with emollients. The scales on the scalp can be cleared with an ointment containing low-concentration sulphur and salicylic acid applied to the scalp daily for a few hours and then washed off. Widespread body eruption will clear with a mild topical corticosteroid, either alone or mixed with an antibacterial and antifungal agent if appropriate.



Figure 24.6 Napkin rash due to *Candida* infection. The skin flexures are involved and there are satellite pustules visible.



(a)



(b)

Figure 24.7 Infantile seborrhoeic dermatitis. **(a)** Cradle cap. **(b)** Involvement of face, axillae and napkin area.

Atopic eczema (atopic dermatitis)

The prevalence of atopic eczema in children in the UK is about 20%. A genetic deficiency of skin barrier function is important in the pathogenesis of atopic eczema. Onset of atopic eczema is usually in the first year of life. It is, however, uncommon in the first 2 months, unlike infantile seborrhoeic dermatitis, which is relatively common at this age. There is often a family history of

atopic disorders: eczema, asthma, allergic rhinitis (hay fever). Around one-third of children with atopic eczema will develop asthma. Exclusive breast-feeding may delay the onset of eczema in predisposed children but does not appear to have a significant impact on the prevalence of eczema during later childhood. Atopic eczema is mainly a disease of childhood, being most severe and troublesome in the first year of life and resolving in 50% by 12 years of age, and in 75% by 16 years.

Diagnosis

The diagnosis is made clinically. If tested, most affected children have an elevated total plasma IgE level. If there is a history to suggest a particular allergic cause, skin-prick and radioallergosorbent (RAST) tests may be helpful. This will also identify food and other allergens which may cause anaphylaxis. If the disease is unusually severe, atypical or associated with unusual infections or failure to thrive, an immune deficiency disorder should be excluded. Immunological changes in atopic disease are probably secondary to enhanced antigen penetration through a deficient epidermal barrier.

Clinical features

Rashes may itch in many conditions (Box 24.2), but in atopic eczema, itching (pruritus) is the main symptom at all ages and this results in scratching and exacerbation of the rash (Fig. 24.8). The excoriated areas become erythematous, weeping and crusted. Distribution of the eruption tends to change with age, as indicated in Figure 24.9.

Atopic skin is usually dry, and prolonged scratching and rubbing of the skin may lead to lichenification, in which there is accentuation of the normal skin markings (Fig. 24.10).

Complications

Causes of exacerbations of eczema are listed in Box 24.3. However, flare-ups are common, often for no obvious reason. Eczematous skin can readily become infected, usually with *Staphylococcus* or *Streptococcus* (Fig. 24.11). Inflammation increases the avidity of skin for *Staph. aureus* and reduces the expression of antimicrobial peptides, which are needed to control microbial infections. *Staph. aureus* thrives on atopic skin and releases superantigens which seem to maintain and worsen eczema. Herpes simplex virus infection, although less frequent, is potentially very serious as it can spread rapidly on atopic skin, causing an extensive vesicular reaction, eczema herpeticum (see Ch. 14). Regional lymphadenopathy is common and often marked in active eczema; it usually resolves when the skin improves.

Management

A number of treatment modalities are available.

Avoiding irritants and precipitants

It is advisable to avoid soap and biological detergents. Clothing next to the skin should be of pure cotton where possible, avoiding nylon and pure woollen garments. Nails need to be cut short to reduce skin damage from scratching, and mittens at night may be helpful in the very young. When an allergen such as cow's milk has been proven to be a precipitant, it should be avoided.

Emollients

These are the mainstay of management, moisturising and softening the skin. They should be applied liberally two or more times a day and after a bath. They include ointments such as one containing equal parts of white

Itching



Figure 24.8 Atopic dermatitis. Inflamed skin worsened by rubbing/scratching. Itch is the key clinical feature in eczema at all ages, leading to an 'itch–scratch–itch' cycle.

Box 24.2 Some itchy rashes

- Atopic eczema
- Chickenpox
- Urticaria/allergic reactions
- Contact dermatitis
- Insect bites/papular urticaria
- Scabies
- Fungal infections
- Pityriasis rosea.

No itch? – then it's not eczema

Eczema

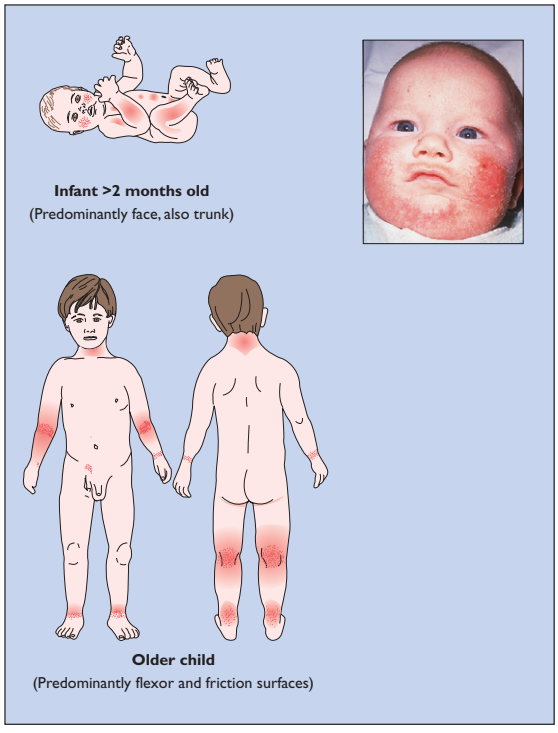


Figure 24.9 Distribution of atopic eczema. The distribution of eczema tends to change with age. In infants, the face and scalp are prominently affected, although the trunk may be involved. In older children, the skin flexures (cubital and popliteal fossae) and frictional areas, such as the neck, wrists and ankles, are characteristically involved.



Figure 24.10 Lichenification.

Box 24.3 Causes of exacerbation of eczema

- Bacterial infection, e.g. *Staphylococcus*, *Streptococcus* spp.
- Viral infection, e.g. herpes simplex virus
- Ingestion of an allergen, e.g. egg
- Contact with an irritant or allergen
- Environment: heat, humidity
- Change or reduction in medication
- Psychological stress
- Unexplained.

Summary

Assessment of the child with eczema

Condition of the skin
Distribution of the eczema: is the skin excoriated, weeping, crusted, lichenified?
How troublesome is the itching?
Worse or better than usual?
What causes exacerbation – food or other allergens, irritants, medications, stress?
Does it disturb sleep?
Does it interfere with life?
Family knowledgeable about condition and its management?

Check:
- Any evidence of infection – bacterial or herpes simplex virus?
- Problems from other allergic disorders?
- Is growth normal?

Management
Avoiding soap, frequently using emollients?
Avoiding nylon and wool clothes?
Is there a need to give or change medications:
- Topical corticosteroids
- Immunomodulators
- Occlusive bandages
- Antibiotics or antiviral agents
- Antihistamines

Allergy test to egg +/- other foods – is it indicated to identify coexistent IgE – mediated food allergy?
On dietary elimination or is it indicated? If so, dietician supervision?
Need for psychosocial support?

Figure 24.11 Infected, excoriated atopic eczema.



soft paraffin and liquid paraffin. Ointments are preferable to creams when the skin is very dry. A daily or alternate day bath using an emollient oil as a soap substitute is also beneficial.

Topical corticosteroids

These are an effective treatment for eczema, but must be used with care. Mildly potent corticosteroids, such as 1% hydrocortisone ointment, can be applied to the eczematous areas twice daily. Moderately potent topical steroids play a pivotal role in the management of acute exacerbations, but their use must be kept to a minimum. They should be applied thinly and their use on the face should be generally avoided. Excessive use of topical steroids may cause thinning of the skin as well as systemic side-effects. However, fear of these side-effects should not deter their use in controlling exacerbations.

Immunomodulators

In children over 2 years old, short-term topical use of tacrolimus or pimecrolimus may be indicated for eczema not controlled by topical corticosteroids and where there is a risk of important adverse effects from further topical steroid use.

Occlusive bandages

These are helpful over limbs when scratching and lichenification are a problem. They may be impregnated with zinc paste or zinc and tar paste. The bandages are worn overnight or for 2–3 days at a time until the skin has improved. For widespread itching in young children, short-term use of wet stockinette wraps may be helpful; diluted topical steroids mixed with emollient are applied to the skin and damp wraps fashioned for trunk and limbs are then applied with overlying dry wraps or clothes.

Antibiotics or antiviral agents

Antibiotics with hydrocortisone can be applied topically for mildly infected eczema. Systemic antibiotics are indicated for more widespread or severe infection. Eczema herpeticum is treated with systemic aciclovir.

Itch suppression is with an oral antihistamine. The newer second-generation antihistamines are not sedative. Antihistamines can be useful in raising the itching threshold so that scratching is reduced.

Dietary elimination

Food allergy may be present if the child reacted with immediate symptoms to a food, or in infants and young children with moderate or severe atopic eczema, particularly if associated with gut dysmotility (colic, vomiting, altered bowel habit) or failure to thrive. It may even occur in young infants with severe eczema who are exclusively breast-fed at the time. The most common food allergies are to egg, cow's milk and peanut. However, any food may be implicated in an eczema flare-up. Dietary elimination for 4–6 weeks is required to detect a response. A trial of an extensively hydrolysed protein formula or amino acid formula in place of cow's milk formula is also recommended for bottle-fed infants under 6 months old with severe atopic eczema that has not been controlled by optimal treatment with emollients and mild topical corticosteroids. Dietary elimination should be carried out with the advice of a dietician to ensure complete avoidance of specific food constituents and that the diet remains nutritionally adequate. A food challenge is required to be fully objective. Children can usually tolerate the offending foods by the age of 3–4 years with the important exception of peanut allergy, which usually persists.

Psychosocial support

In most children, eczema is mild and can be controlled with emollients and mildly potent topical steroids, and additional psychological support is not required. However, eczema can be sufficiently severe to be disrupting both to the child and to the whole family. The parents and the child need considerable advice, help and support from health professionals, other affected families or fellow sufferers. In the UK, the National Eczema Society provides support and education about the disorder.

Infections and infestations

Bullous impetigo has been considered earlier in this chapter and acute bacterial and viral infections of the skin are considered in Chapter 14.

Viral infections

Viral warts

These are caused by the human papillomavirus, of which there are well over 100 types. Warts are common in children, usually on the fingers and soles (verrucae). Most disappear spontaneously over a few months or years and treatment is only indicated if the lesions are painful or are a cosmetic problem. They can be difficult to treat, but daily application of a proprietary salicylic acid and lactic acid paint or glutaraldehyde (10%) lotion can be used. Cryotherapy with liquid nitrogen is effective treatment but can be painful and often needs repeated application, and its use should be reserved for older children.

Molluscum contagiosum

This is caused by a poxvirus. The lesions are small, skin-coloured, pearly papules with central umbilication (Fig. 24.12). They may be single but are usually multiple. Lesions are often widespread but tend to disappear spontaneously within a year. If necessary, a topical antibacterial can be applied to prevent or treat secondary bacterial infection, and cryotherapy (2–3 s only) can be used in older children, away from the face, to hasten the disappearance of more chronic lesions.

Fungal infections

Ringworm

Dermatophyte fungi invade dead keratinous structures, such as the horny layer of skin, nails and hair. The term 'ringworm' is used because of the often ringed



Figure 24.12 Molluscum contagiosum. Some of the pearly lesions show characteristic umbilication.



Figure 24.13 Ringworm of the scalp showing hair loss and kerion.

(annular) appearance of skin lesions. A severe inflammatory pustular ringworm patch is called a kerion (Fig. 24.13).

Tinea capitis (scalp ringworm), sometimes acquired from dogs and cats, causes scaling and patchy alopecia with broken hairs. Examination under filtered ultraviolet (Wood's) light may show bright greenish/yellow fluorescence of the infected hairs with some fungal species.

Rapid diagnosis can be made by microscopic examination of skin scrapings for fungal hyphae. Definitive identification of the fungus is by culture. Treatment of mild infections is with topical antifungal preparations, but more severe infections require systemic antifungal treatment for several weeks. Any animal source of infection also needs to be treated.

Summary

Tinea capitis (scalp ringworm)

- Annular scaling scalp lesion with patchy alopecia with broken hairs
- Fungal hyphae on skin scrapings
- Treated with topical or systemic antifungal
- Treat the dog or cat, if infected.

Parasitic infestations

Scabies

Scabies is caused by an infestation with the eight-legged mite *Sarcoptes scabiei*, which burrows down the epidermis along the stratum corneum. Severe itching occurs 2–6 weeks after infestation and is worse in warm conditions and at night.

In older children, burrows, papules and vesicles involve the skin between the fingers and toes, axillae, flexor aspects of the wrists, belt line and around the nipples, penis and buttocks. In infants and young children, the distribution often includes the palms (Fig. 24.14), soles and trunk. The presence of lesions on the soles can be helpful in making the diagnosis. The head, neck and face can be involved in babies but is uncommon.



Figure 24.14 Scabies in a young child affecting the palm.

Diagnosis is made on clinical grounds with the history of itching and characteristic lesions. Although burrows are considered pathognomonic, they may be hard to identify because of secondary infection due to scratching. Itching in other family members is a helpful clinical indicator. Confirmation can be made by microscopic examination of skin scrapings from the lesions to identify mite, eggs and mite faeces.

Complications

The skin becomes excoriated due to scratching and there may be a secondary eczematous or urticarial reaction masking the true diagnosis. Secondary bacterial infection is common, giving crusted, pustular lesions. Sometimes slowly resolving nodular lesions are visible.

Treatment

As it is spread by close bodily contact, the child and whole family should be treated, whether or not they have evidence of infestation. Permethrin cream (5%) should be applied below the neck to all areas and washed off after 8–12 h. In babies, the face and scalp should be included, avoiding the eyes. Benzyl benzoate emulsion (25%) applied below the neck only, in diluted form according to age, and left on for 12 h, is also effective but smells and has an irritant action. Malathion lotion (0.5% aqueous) is another effective preparation applied below the neck and left on for 12 h.

If a child and other members of the family are itching, suspect scabies.

Summary

Scabies

- Very itchy burrows, papules and vesicles – distribution varies with age
- Scratching leads to excoriation, secondary eczematous or urticarial reaction often with secondary bacterial infection
- Not only the child but the whole family need treatment.

Pediculosis

Pediculosis capitis (head lice infestation) is the most common form of lice infestation in children. It is widespread and troublesome among primary school children. Presentation may be itching of the scalp and nape or from identifying live lice on the scalp or nits (empty egg cases) on hairs (Fig. 24.15). Louse eggs are cemented to hair close to the scalp and the nits (small whitish oval capsules) remain attached to the hair shaft as the hair grows. There may be secondary bacterial infection, often over the nape of the neck, leading to a misdiagnosis of impetigo. Sub-occipital lymphadenopathy is common. Once infestation is confirmed by finding live lice, treatment is by applying a solution of 0.5% malathion to the hair and leaving it on overnight. The hair is then shampooed and the lice and nits removed with a fine-tooth comb. Treatment should be repeated a week later. Permethrin (1%) as a cream rinse would be an alternative application; it is left on for 10 min only. Flammability of alcohol-based lotions should be noted. Wet combing to remove live lice (bug-busting) every 3–4 days for at least 2 weeks is a useful and safe physical treatment, particularly when parents treat with enthusiasm.



Figure 24.15 Head lice. Profuse nits (egg capsules) are visible on scalp hairs. Live lice were visible on the scalp.

Other childhood skin disorders

Psoriasis

This familial disorder rarely presents before the age of 2 years. The guttate type (Fig. 24.16) is common in children and often follows a streptococcal or viral sore throat or ear infection. Lesions are small, raindrop-like, round or oval erythematous scaly patches on the trunk and upper limbs, and an attack usually resolves over 3–4 months. However, most get a recurrence of psoriasis within the next 3–5 years. Chronic psoriasis with plaques or annular lesions is less common. Fine pitting of the nails may be seen in chronic disease but is unusual in children. Treatment for guttate psoriasis is with bland ointments. Coal tar preparations are useful for plaque psoriasis and scalp involvement. Dithranol preparations are very effective in resistant plaque psoriasis. Calcipotriol, a vitamin D analogue, which does not stain the skin, can also be useful for plaque psoriasis in those over 6 years old. Occasionally, children with chronic psoriasis develop psoriatic arthritis. Chronic psoriasis may have a considerable effect on quality of life. The Psoriasis Association can be helpful in offering support and advice.



Figure 24.16 Guttate psoriasis over the back in a 5-year-old.

Pityriasis rosea

This acute, benign self-limiting condition is thought to be of viral origin. It usually begins with a single round or oval scaly macule, the herald patch, 2–5 cm in diameter, on the trunk, upper arm, neck or thigh. After a few days, numerous smaller dull pink macules develop on the trunk, upper arms and thighs. The rash tends to follow the line of the ribs posteriorly, described as the 'fir tree pattern'. Sometimes the lesions are itchy.



Figure 24.17 Alopecia areata. Smooth well-defined patch of non-inflamed hair fall.

No treatment is required and the rash resolves within 4–6 weeks.

Alopecia areata

This is a common form of hair loss in children and, understandably, a cause of much family distress. Hairless, single or multiple non-inflamed smooth areas of skin, usually over the scalp, are present (Fig. 24.17); remnants of broken-off hairs, visible as 'exclamation mark' hairs may be seen at the edge of active patches of hair fall. The more extensive the hair loss, the poorer the prognosis, but regrowth often occurs within 6–12 months in localised hair loss. Prognosis should be more guarded in children with atopic disorders.

Granuloma annulare

Lesions are typically ringed (annular) with a raised flesh-coloured non-scaling edge (unlike ringworm) (Fig. 24.18). They may occur anywhere but usually over bony prominences, especially over hands and feet.



Figure 24.18 Granuloma annulare. Ringed lesion with a non-inflamed, non-scaling raised edge.

Lesions may be single or multiple, are usually 1–3 cm in diameter, and tend to disappear spontaneously but may take years to do so. There is also a subcutaneous form.

Acne vulgaris

Acne may begin 1–2 years before the onset of puberty following androgenic stimulation of the sebaceous glands and an increased sebum excretion rate. Obstruction to the flow of sebum in the sebaceous follicle initiates the process of acne. There are a variety of lesions, initially open comedones (blackheads) or closed comedones (whiteheads) progressing to papules, pustules, nodules and cysts. Lesions occur mainly on the face, back, chest and shoulders. The more severe cystic and nodular lesions often produce scarring. Menstruation and emotional stress may be associated with exacerbations. The condition usually resolves in the late teens, although it may persist.

Topical treatment is directed at encouraging the skin to peel using a keratolytic agent, such as benzoyl peroxide, applied once or twice daily after washing. Sunshine, in moderation, topical antibiotics or topical retinoids may be helpful. For more severe acne, oral antibiotic therapy with tetracyclines (only when over 12 years old, because they may discolour the teeth in younger children) or erythromycin is indicated. The oral retinoid isotretinoin is reserved for severe acne in teenagers unresponsive to other treatments.

Rashes and systemic disease

Skin rashes may be a sign of systemic disease. Examples are:

- Facial rash in systemic lupus erythematosus (SLE) or dermatomyositis
- Purpura over the buttocks, lower limbs and elbows in Henoch–Schönlein purpura
- Erythema nodosum (Fig. 24.19, Box 24.4) and erythema multiforme (Fig. 24.20, Box 24.5); both can be associated with a systemic disorder, but often no cause is identified
- Stevens–Johnson syndrome, a severe bullous form of erythema multiforme also involving the mucous membranes (Fig. 24.21). The eye involvement may include conjunctivitis, corneal ulceration and uveitis, and ophthalmological assessment is required. It may be caused by drug sensitivity or infection, with morbidity and sometimes even mortality from infection, toxaemia or renal damage
- Urticaria.

Urticaria

Urticaria (hives), characterised by flesh-coloured weals, is described in Chapter 15 and the management of anaphylaxis in Chapter 6.

Papular urticaria is a delayed hypersensitivity reaction most commonly seen on the legs, following a bite from a flea, bedbug, or animal or bird mite. Irritation,

Erythema nodosum

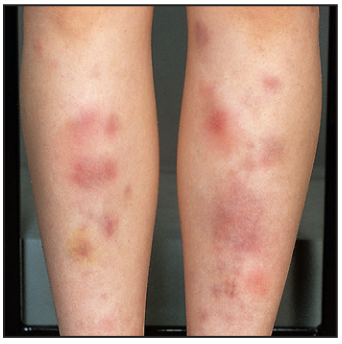


Figure 24.19 Erythema nodosum. There are tender nodules over the legs. She also had fever and arthralgia.

Box 24.4 Causes of erythema nodosum

- Streptococcal infection
- Primary tuberculosis
- Inflammatory bowel disease
- Drug reaction
- Idiopathic

(Sarcoidosis, a common association in adults, is rare in children)

Erythema multiforme



Box 24.5 Causes of erythema multiforme

- Herpes simplex infection
- *Mycoplasma pneumoniae* infection
- Other infections
- Drug reaction
- Idiopathic

Figure 24.20 There are target lesions with a central papule surrounded by an erythematous ring. Lesions may also be vesicular or bullous.



Figure 24.21 Stevens–Johnson syndrome showing severe conjunctivitis and ulceration of the mouth. (Courtesy of Dr Rob Primhak.)

vesicles, papules and weals appear and secondary infection due to scratching is common. It may last for weeks or months and may be recurrent.

Hereditary angioedema is a rare autosomal dominant disorder caused by a deficiency or dysfunction of C1-esterase inhibitor. There is no urticaria, but subcutaneous swellings occur, often accompanied by abdominal pain. The trigger is usually trauma. Angioedema may cause respiratory obstruction. Specific treatment of a severe acute attack is with a purified preparation of the inhibitor, but replacement therapy with fresh frozen plasma can be used as a short-term measure.

Further reading

Chiang NY, Verbov J: *Dermatology – A Handbook for Medical Students and Junior Doctors*, London, 2009, British Association of Dermatologists. *Available free from the British Association of Dermatologists, London*

25

Endocrine and metabolic disorders

Diabetes mellitus	433	Cushing syndrome	445
Hypoglycaemia	441	Inborn errors of metabolism	446
Hypothyroidism	442	Newborn screening	446
Hyperthyroidism	443	Disorders presenting acutely in the neonatal period	447
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Adrenal cortical insufficiency	445		

Points of note concerning endocrine and metabolic disorders in children are:

- The number of children developing diabetes mellitus is increasing
- The most common cause of hypothyroidism is congenital, which is detected on routine biochemical screening shortly after birth
- Inborn errors of metabolism are individually very rare and managed by a few specialist centres.

Diabetes mellitus

The incidence of diabetes in children has increased steadily over the last 20 years and now affects around 2 per 1000 children by 16 years of age. It has been estimated that the incidence of childhood diabetes will double by 2020 in developed countries. This is most likely to be a result of changes in environmental risk factors, although the exact causes remain obscure. There is considerable racial and geographical variation – the condition is more common in northern countries, with high incidences in Scotland and Finland. Almost all children have type 1 diabetes requiring insulin from the outset. Type 2 diabetes due to insulin resistance is starting to occur in childhood, as severe obesity becomes more common and in some ethnic groups. The other causes of diabetes are listed in Box 25.1.

Almost all children with diabetes have insulin-dependent (type 1) diabetes, although type 2 diabetes is increasingly common. In 2009, there were approximately 25 000 children and young people with type 1 diabetes and 200 with type 2 diabetes in the UK.

Aetiology of type 1 diabetes

Both genetic predisposition and environmental precipitants play a role. Inherited susceptibility is demonstrated by:

- An identical twin of a diabetic has a 30–40% chance of developing the disease

Box 25.1 Classification of diabetes according to aetiology

- **Type 1. Most childhood diabetes:**
  - Destruction of pancreatic β-cells by an autoimmune process
- **Type 2. Insulin resistance followed later by β-cell failure:**
  - Usually older children, obesity-related, positive family history, not as prone to ketosis, commoner in some ethnic groups (e.g. Indian subcontinent)
- **Type 3. Other specific types:**
  - Genetic defects in β-cell function (maturity-onset diabetes of the young, MODY) due to glucokinase or transcription factor mutations
  - Genetic defects in insulin action
  - Infections, e.g. congenital rubella
  - Drugs, e.g. corticosteroids
  - Pancreatic exocrine insufficiency, e.g. cystic fibrosis
  - Endocrine diseases, e.g. Cushing syndrome
  - Genetic/chromosomal syndromes, e.g. Down and Turner
  - Neonatal diabetes: transient and permanent
- **Type 4. Gestational diabetes (GDM).**

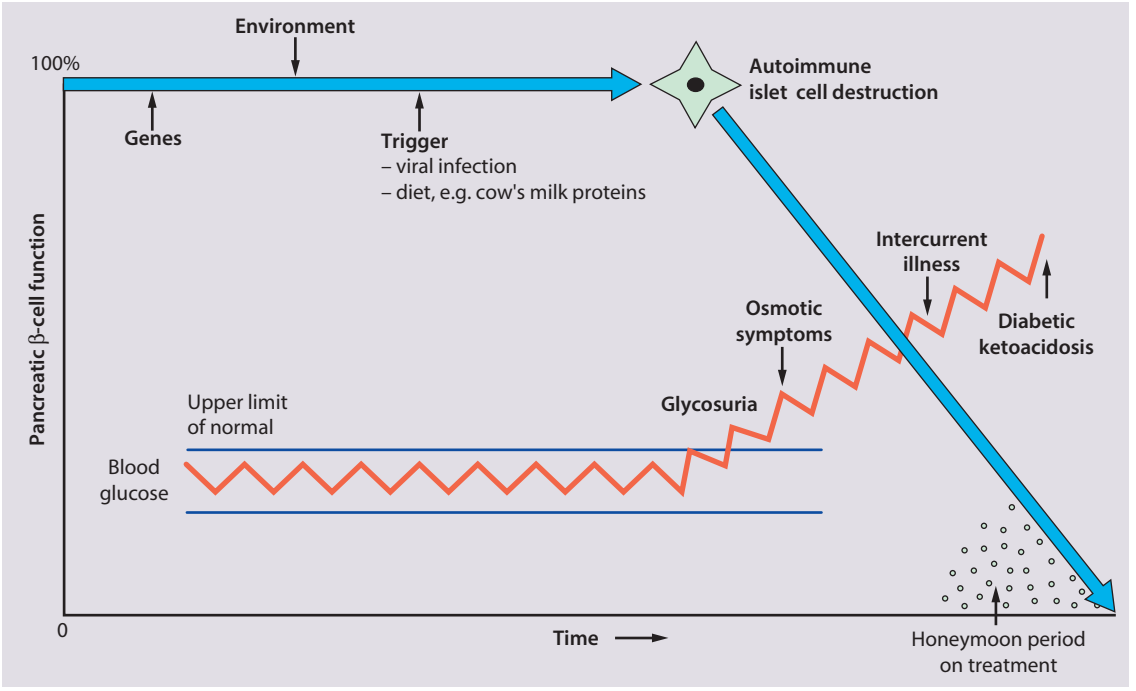


Figure 25.1 Stages in the development of diabetes.

- The increased risk of a child developing diabetes if a parent has insulin-dependent diabetes (1 in 20–40 if the father is affected, 1 in 40–80 if it is the mother – compared to about 1/400 in the population <16 years)
- The increased risk of diabetes among those who are HLA-DR3 or HLA-DR4 and a reduced risk with DR2 and DR5.

Molecular mimicry probably occurs between an environmental trigger and an antigen on the surface of β-cells of the pancreas. Triggers which may contribute are enteroviral infections, accounting for the more frequent presentation in spring and autumn, and diet, possibly cow's milk proteins (Fig. 25.1) and overnutrition. In genetically predisposed individuals, this results in an autoimmune process which damages the pancreatic β-cells and leads to increasing insulin deficiency. Markers of β-cell destruction include islet cell antibodies and antibodies to glutamic acid decarboxylase (GAD), the islet cells and insulin. There is an association with other autoimmune disorders such as hypothyroidism, Addison disease, coeliac disease and rheumatoid arthritis in the patient or family history.

Clinical features

There are two peaks of presentation of type 1 diabetes, preschool and teenagers. It is also commoner to present in spring and autumn months. In contrast to adults, children usually present with only a few weeks of polyuria, excessive thirst (polydipsia) and weight loss; young children may also develop secondary nocturnal enuresis. Most children are diagnosed at this early stage of the illness (Box 25.2). Advanced diabetic ketoacidosis has become an uncommon presentation (<10% in some areas of the UK), but requires urgent recognition and treatment. Diabetic ketoacidosis may be misdiagnosed if the hyperventilation is mistaken for pneumonia or the abdominal pain for appendicitis or constipation.

Diagnosis

The diagnosis is usually confirmed in a symptomatic child by finding a markedly raised random blood

Box 25.2 Symptoms and signs of diabetes

Early

- *Most common – the 'classical triad':*
  - Excessive drinking (polydipsia)
  - Polyuria
  - Weight loss
- *Less common:*
  - Enuresis (secondary)
  - Skin sepsis
  - *Candida* and other infections

Late – diabetic ketoacidosis

- Smell of acetone on breath
- Vomiting
- Dehydration
- Abdominal pain
- Hyperventilation due to acidosis (Kussmaul breathing)
- Hypovolaemic shock
- Drowsiness
- Coma and death.



Figure 25.2 Acanthosis nigricans in axilla. A sign of insulin resistance.

glucose (>11.1 mmol/L by the current WHO definition), glycosuria and ketonuria. Where there is any doubt, a fasting blood glucose (>7 mmol/L) or a raised glycosylated haemoglobin (HbA_{1c}) are helpful. A diagnostic glucose tolerance test is rarely required in children.

Type 2 diabetes should be suspected if there is a family history, in children from the Indian subcontinent and in severely obese children with signs of insulin resistance (acanthosis nigricans – velvety dark skin on the neck or armpits (Fig. 25.2), skin tags or the polycystic ovary phenotype in teenage girls).

Initial management of type 1 diabetes

As type 1 diabetes in childhood is uncommon (1–2 children per large secondary school), much of the initial and routine care is delivered by specialist teams (Box 25.3).

The initial management will depend on the child's clinical condition. Those in advanced diabetic ketoacidosis require urgent hospital admission and treatment (see below). Most newly presenting children are alert and able to eat and drink and can be managed with subcutaneous insulin alone. Intravenous fluid is required if the child is vomiting or dehydrated. In most centres with sufficient resources, children newly presenting with diabetes who do not require intravenous therapy are not admitted to hospital but are managed entirely at home.

An intensive educational programme is needed for the parents and child, which covers:

- A basic understanding of the pathophysiology of diabetes
- Injection of insulin: technique and sites
- Diet: reduced refined carbohydrate; healthy diet with no more than 30% fat intake; 'carbohydrate counting', estimating the amount of carbohydrate in food to allow calculation of the insulin required for each meal or snack
- Adjustments of diet and insulin for exercise
- 'Sick-day rules' during illness to prevent ketoacidosis
- Blood glucose (finger prick) monitoring and blood ketones when unwell

Box 25.3 The diabetes team

- Consultant paediatrician(s) with a special interest in diabetes
- Paediatric diabetes specialist nurse(s)
- Paediatric dietician
- Clinical psychologist
- Social worker
- Adult diabetologist for joint adolescent clinics
- Parent/patient support groups.

- The recognition and staged treatment of hypoglycaemia
- Where to get advice 24 hours a day
- The help available from voluntary groups, e.g. local groups or 'Diabetes UK'
- The psychological impact of a lifelong condition with potentially serious short- and long-term complications.

A considerable period of time needs to be spent with the family to provide this information and psychological support. The information provided for the child must be appropriate for age, and updated regularly. The specialist nurse should liaise with the school (teachers, those who prepare school meals, physical education teachers) and the primary care team.

Insulin

Insulin is made chemically identical to human insulin by recombinant DNA technology or by chemical modification of pork insulin. All insulin that is used in the UK in children is human and in concentrations of 100 U/ml (U-100). The types of insulin include:

- Human insulin analogues. Rapid-acting insulin analogues, e.g. insulin lispro, insulin glulisine or insulin aspart (trade names Humalog, Apidra and NovoRapid, respectively) – with a much faster onset and shorter duration of action than soluble regular insulin. There are also very long-acting insulin analogues, e.g. insulin detemir (Levemir) or glargine (Lantus)
- 'Short-acting' soluble human regular insulin. Onset of action (30–60 min), peak 2–4 h, duration up to 8 h. Given 15–30 min before meals. Trade named examples are Actrapid and Humulin S
- Intermediate-acting insulin. Onset 1–2 h, peak 4–12 h. Isophane insulin is insulin with protamine, e.g. Insulatard and Humulin I
- Predetermined preparations of mixed short- and intermediate-acting insulins with 25% or 30% rapid-acting components.

Insulin can be given by continuous infusion of rapid-acting insulin from a pump or by injections using a variety of syringe and needle sizes, pen-like devices with insulin-containing cartridges, and jet injectors that inject insulin needle-free as a fine stream into the subcutaneous tissue.

Insulin may be injected into the subcutaneous tissue of the upper arm, the anterior and lateral aspects of the thigh, the buttocks and the abdomen. Rotation